



Power Electronics Project

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Contents

1	Introduction	2
2	Square wave conduction	3
3	Quasi-square wave output that eliminates the third harmonic:	7
4	Selected Harmonic Elimination (SHE)	11
4.1	MATLAB Implementation of SHE Switching Angle Calculation	12
4.2	Numerical Results	14
5	Bipolar and Unipolar PWM Modulation Techniques	18
5.1	Unipolar PWM	18
5.2	Bipolar PWM Simulation Overview	19
5.2.1	Bipolar PWM Simulation:	20
5.3	Unipolar PWM Simulation Overview	24
6	Bipolar and Unipolar PWM find the minimum value of mf	30
6.1	unipolar calculations:	31
6.2	Bipolar calculations:	33
7	Multi-level inverter	37

Abstract

This report presents the analysis, modeling, and simulation of a single-phase inverter using MATLAB/Simulink. The inverter is designed to operate with a DC input voltage of 20 V and an R–L load consisting of 34 Ω resistance and 33 mH inductance at a 50 Hz output frequency. Multiple modulation strategies are implemented and evaluated, including square wave conduction, quasi-square wave operation with third-harmonic elimination, selective harmonic elimination (SHE) targeting the removal of the 3rd and 5th harmonics, and both bipolar and unipolar PWM techniques. Additional analysis is performed to determine the minimum switching frequency modulation index that satisfies the requirements of $V_{1,\text{rms}} = \frac{16}{\sqrt{2}}$ and a total harmonic distortion (THD) of 10% or less. A multilevel inverter configuration is also developed as an extended task.

For each modulation technique, waveforms of gate signals, load voltage, load current, and supply current are obtained and analyzed. The results are compared in terms of harmonic distortion and output waveform quality, and a summary table is provided to present the performance of all techniques. The study demonstrates the effectiveness of optimized switching strategies in improving inverter output quality while satisfying power quality constraints.

1. Introduction

Power electronic inverters play a critical role in modern energy conversion systems, enabling the transformation of DC power into AC power for a wide range of applications including renewable energy systems, industrial drives, and consumer electronics. The performance of an inverter is largely determined by the switching strategy used to control the output waveform, as different modulation techniques yield different levels of harmonic distortion, switching losses, and output quality.

This project focuses on the simulation and analysis of a single-phase inverter using MATLAB/Simulink, with the objective of studying how various modulation methods influence inverter performance. The inverter under investigation operates with a DC input of 20 V and supplies an R–L load designed to emulate practical electrical characteristics. Several modulation techniques are explored, beginning with basic square-wave conduction and progressing toward more advanced approaches such as quasi-square wave generation, selective harmonic elimination (SHE), and high-frequency bipolar and unipolar pulse-width modulation (PWM). Each technique is assessed based on its ability to reduce undesirable harmonic components, achieve the required fundamental voltage level, and maintain acceptable THD values.

Furthermore, the study examines the minimum necessary switching frequency for

PWM strategies to meet specified performance criteria. A multilevel inverter topology is also implemented as a bonus enhancement to investigate the benefits of increased voltage levels on harmonic performance.

Through waveform analysis, THD measurements, and systematic comparison across modulation methods, this project provides a comprehensive understanding of single-phase inverter behavior and highlights the trade-offs associated with different switching techniques. The results support the selection of modulation strategies that enhance power quality and optimize inverter efficiency.

2. Square wave conduction

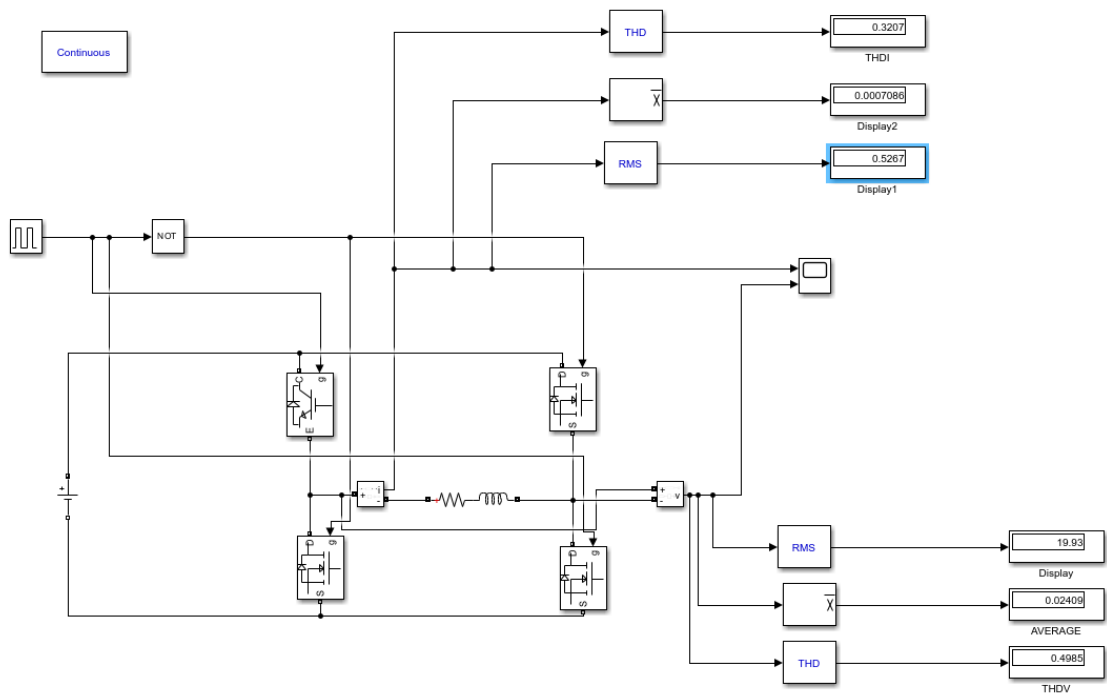


Figure 1: Square wave model

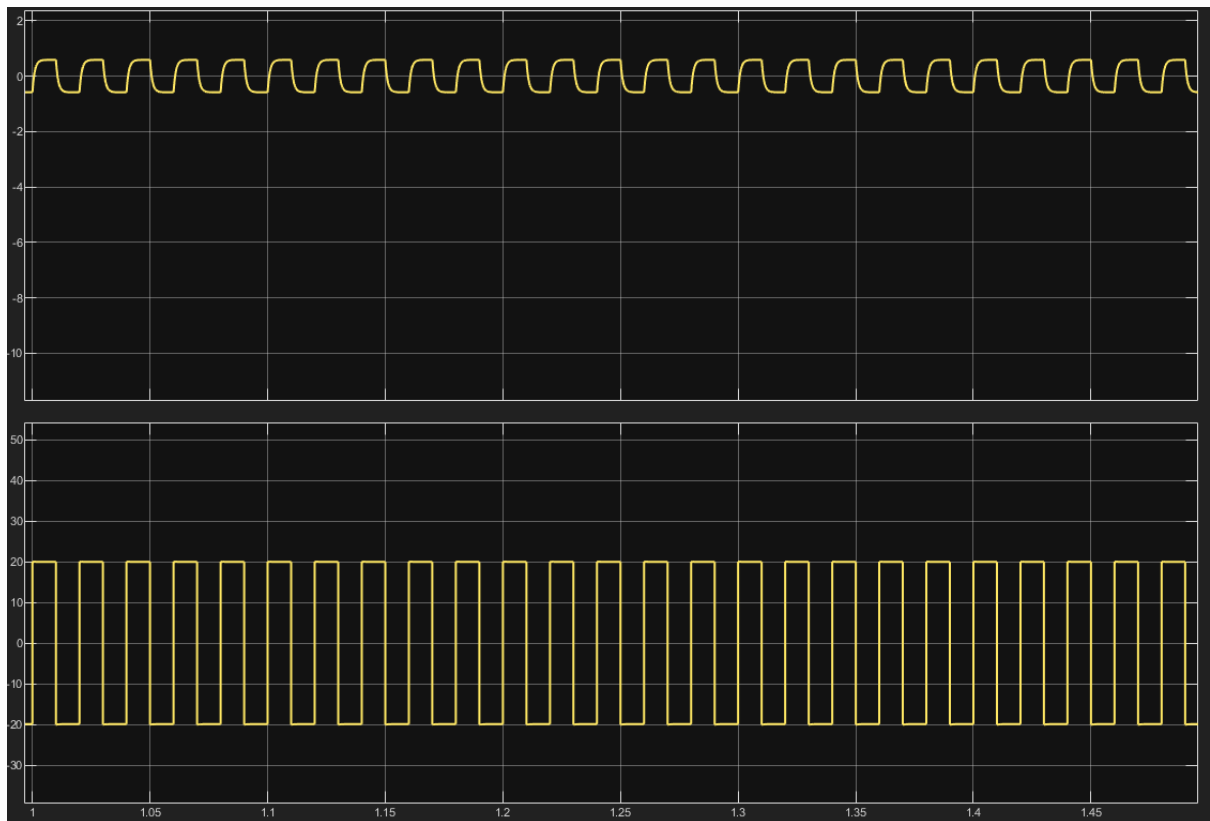


Figure 2: wave forms

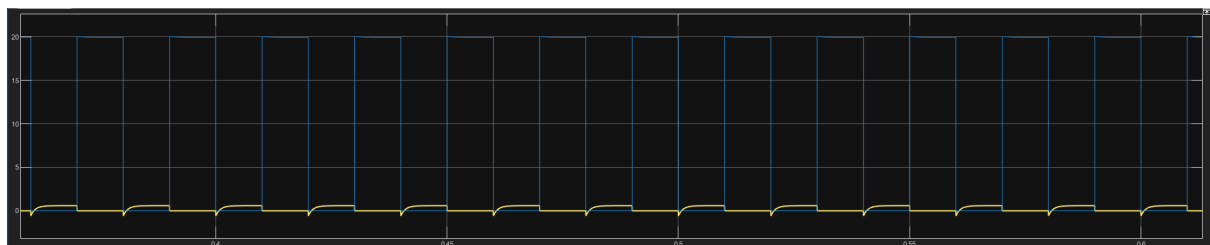


Figure 3: s1,s2

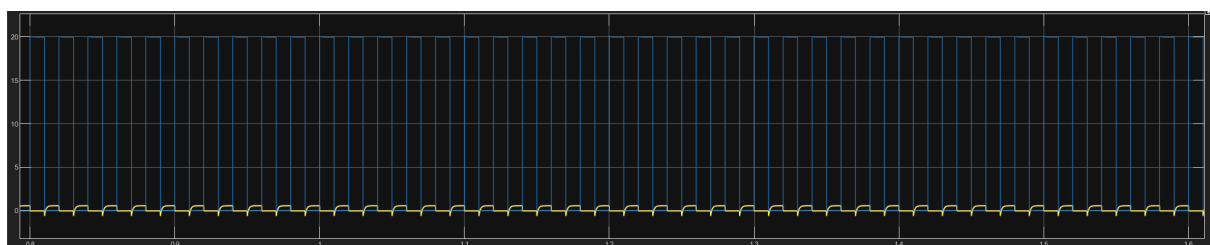


Figure 4: s3,s4

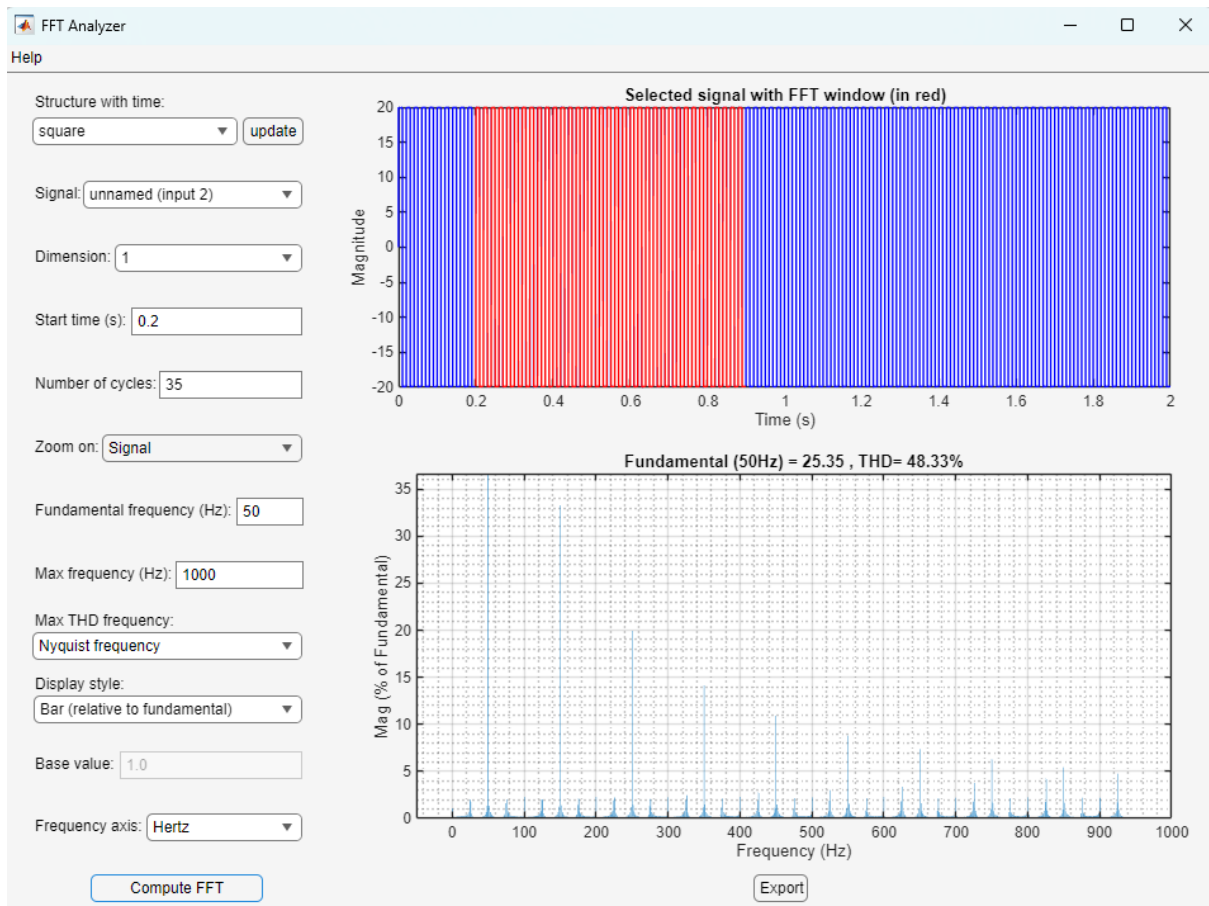


Figure 5: FFT voltage

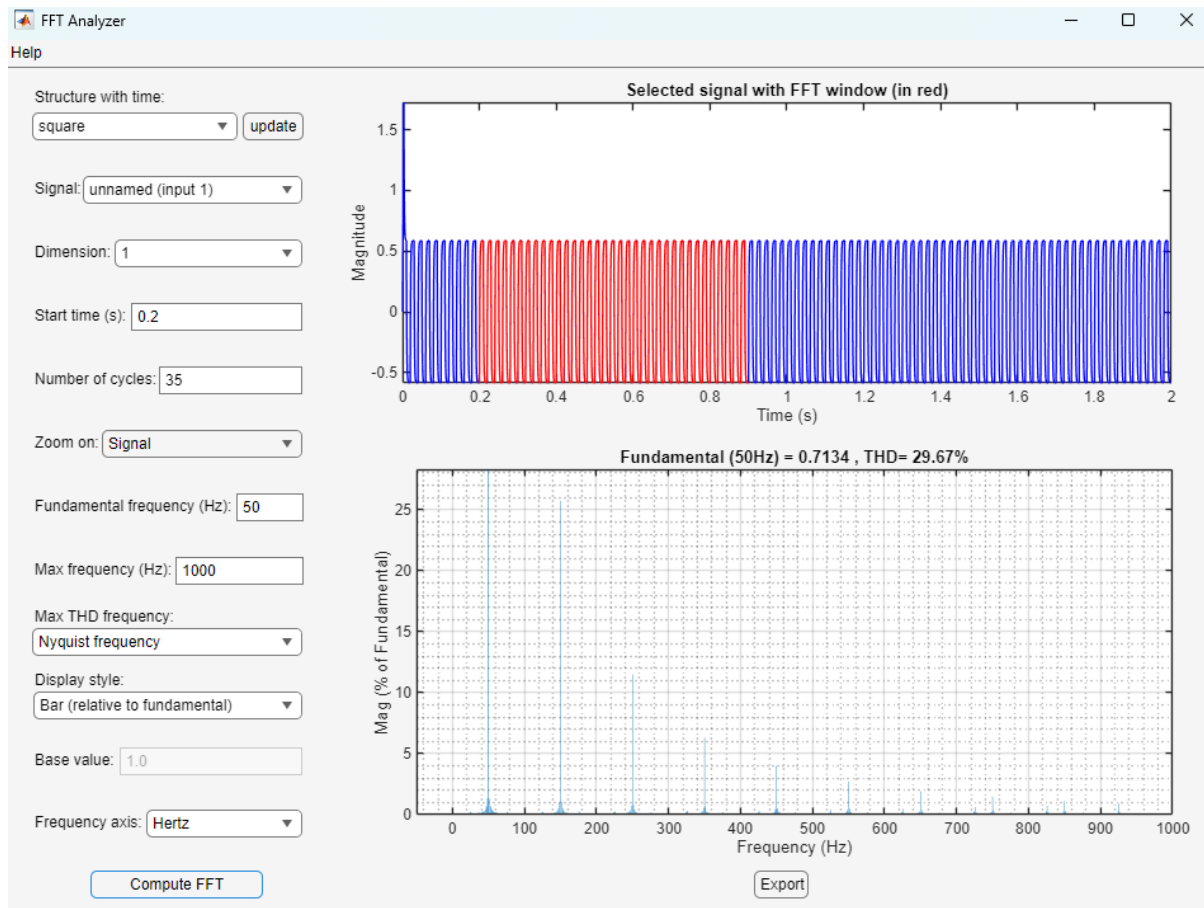


Figure 6: FFT current

3. Quasi-square wave output that eliminates the third harmonic:

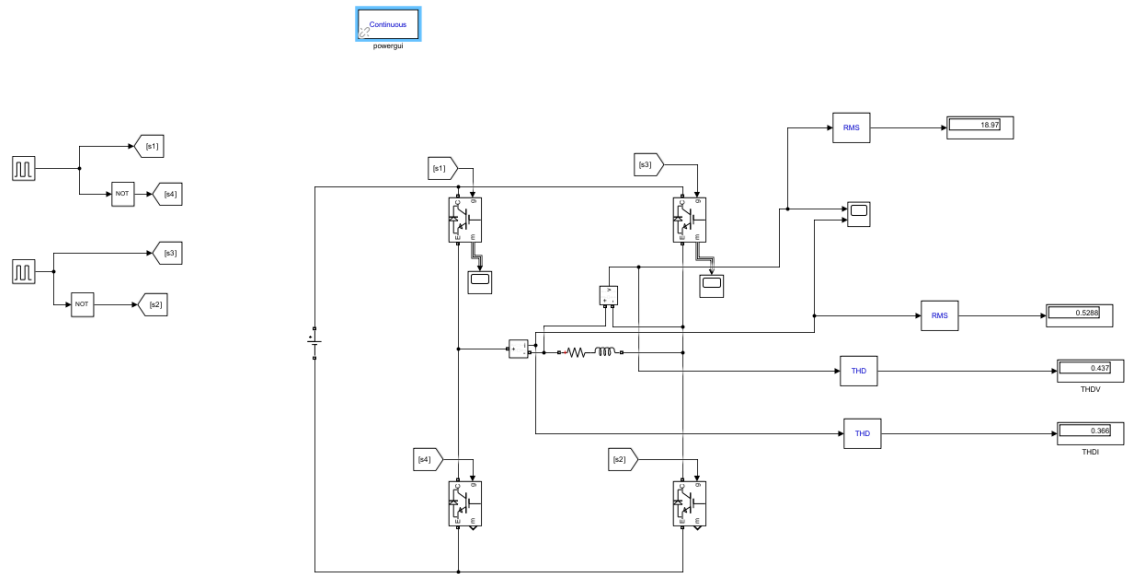


Figure 7: Quasi-square wave model

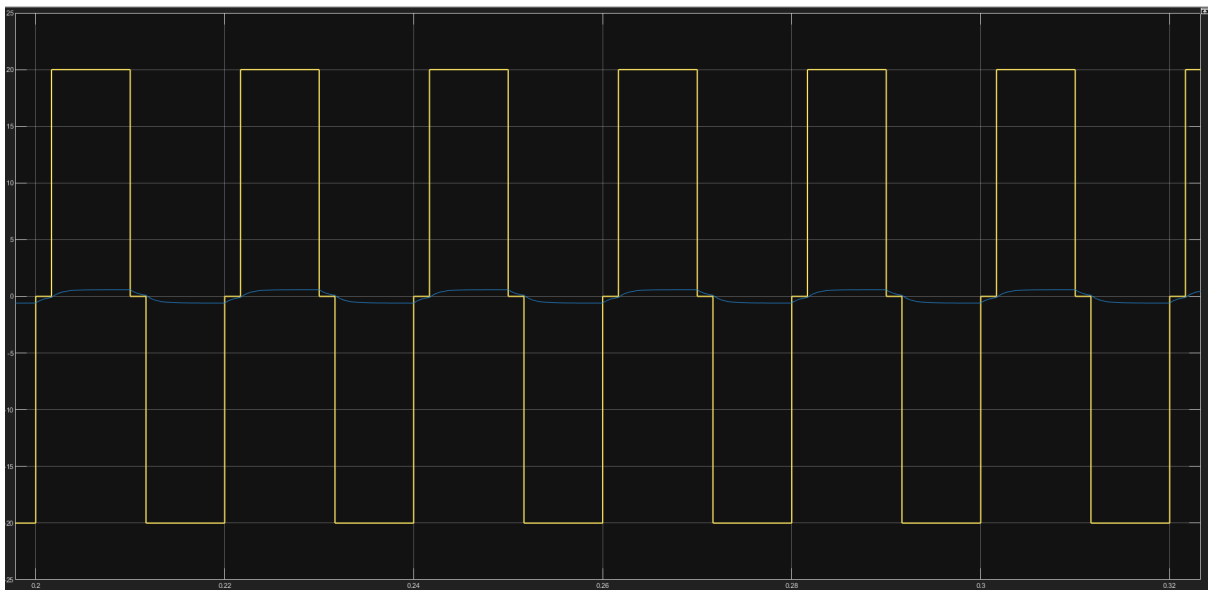


Figure 8: Quasi-square wave wave forms

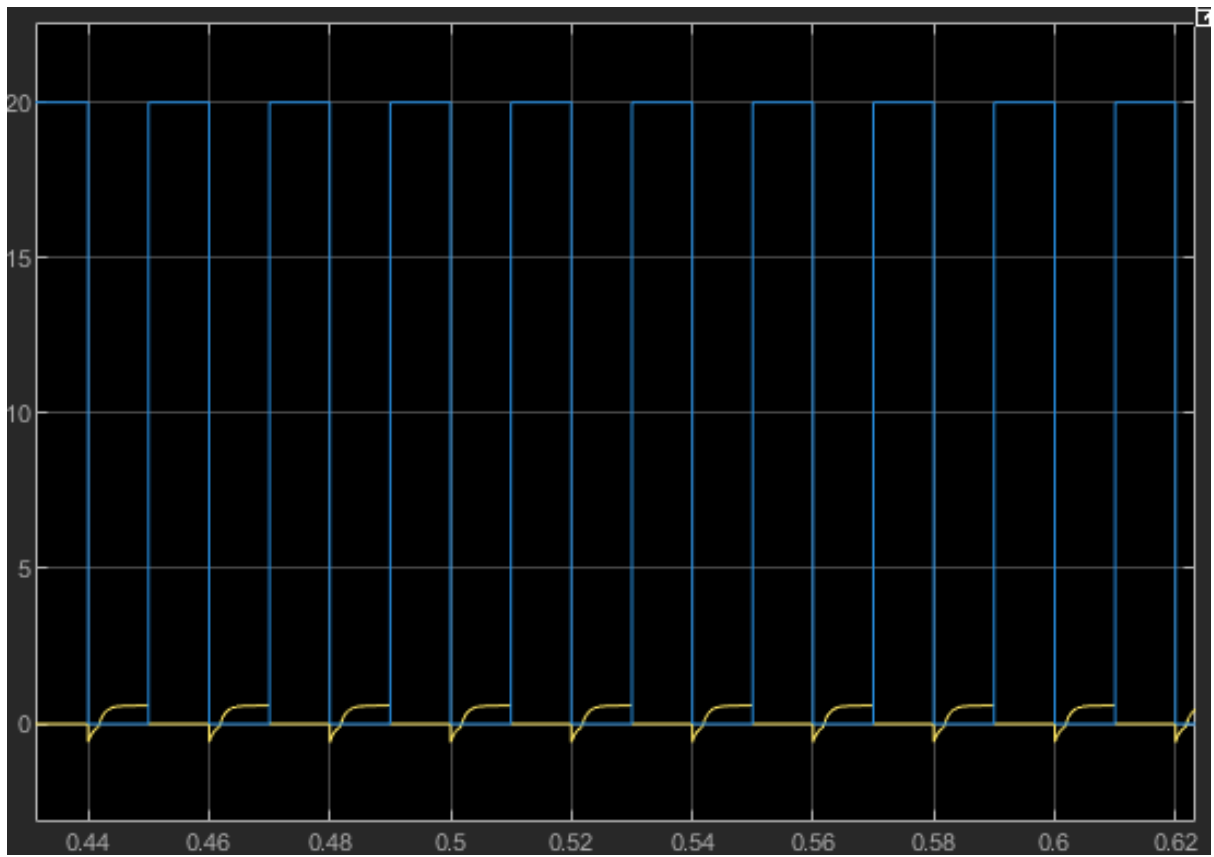


Figure 9: s_1, s_2

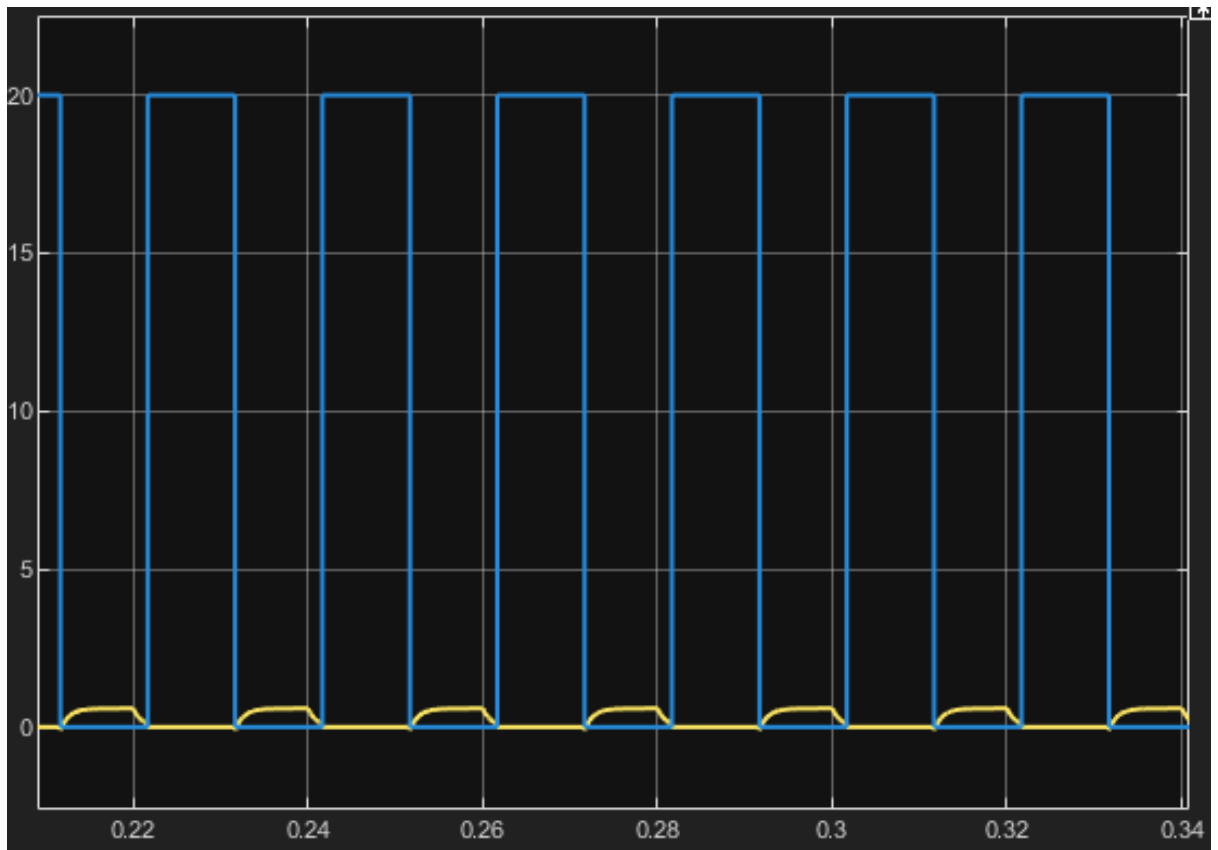


Figure 10: s_3, s_4

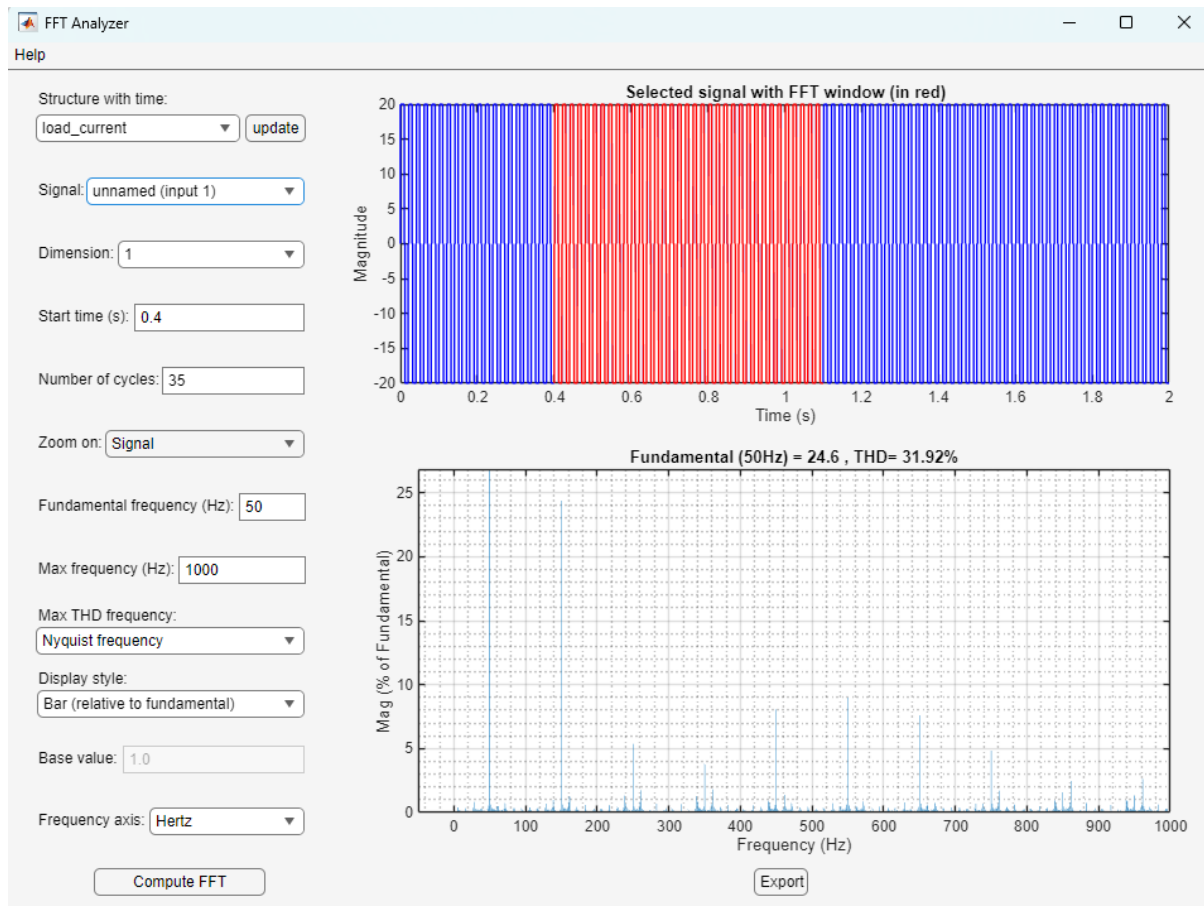


Figure 11: FFT voltage

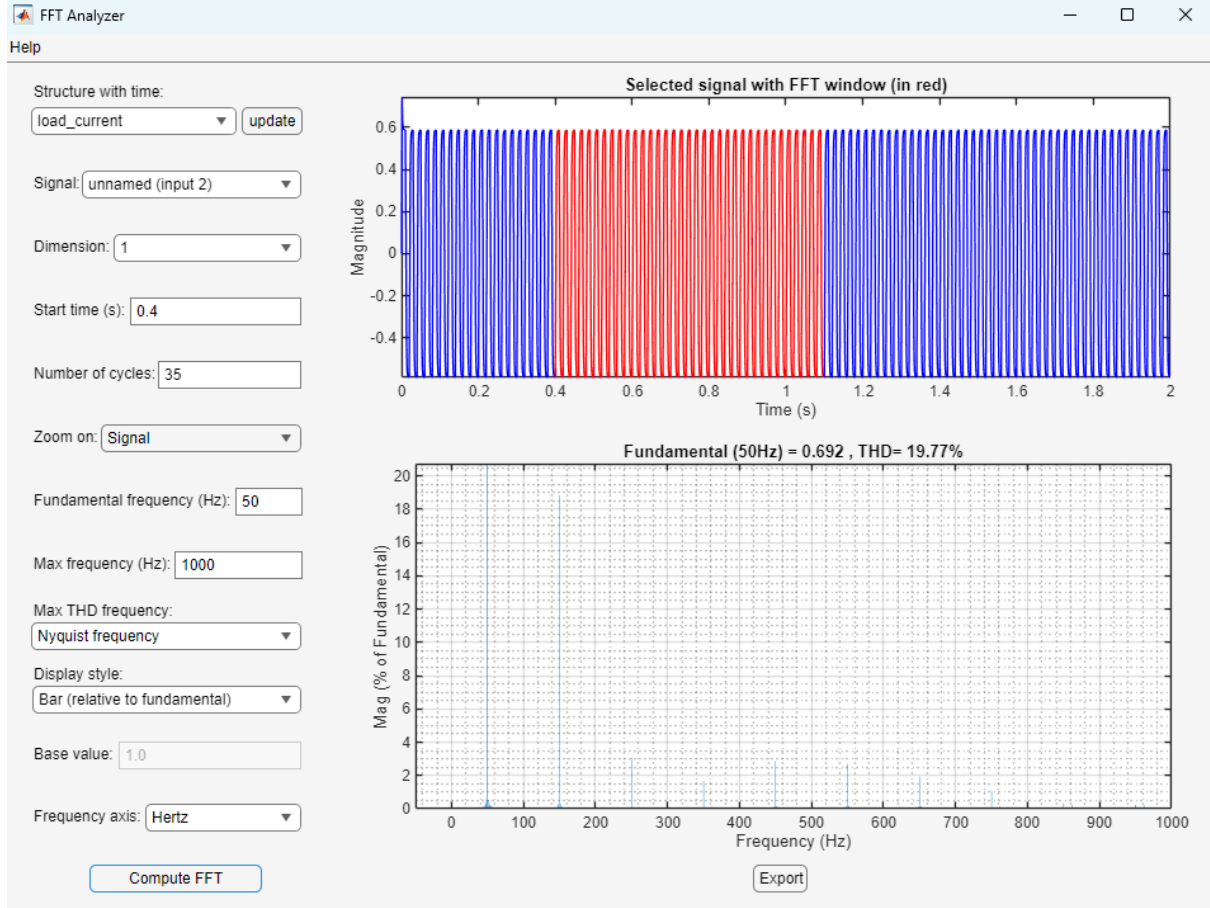


Figure 12: FFT current

4. Selected Harmonic Elimination (SHE)

In this part of the project, the selected harmonic elimination (SHE) technique is applied to the single-phase full-bridge inverter in order to cancel the 3rd and 5th voltage harmonics while maintaining a prescribed fundamental component. The inverter output is assumed to have half-wave and quarter-wave symmetry, so only odd sine harmonics are present. The waveform in each quarter cycle is defined by three switching angles $0 < \alpha_1 < \alpha_2 < \alpha_3 < 90^\circ$.

Using Fourier series, the n th harmonic coefficient of the line voltage is written as

$$b_n = \frac{4V_{dc}}{n\pi} [\cos(n\alpha_1) - \cos(n\alpha_2) + \cos(n\alpha_3)].$$

The SHE problem is formulated by imposing three conditions on these coefficients. First, the fundamental harmonic equation:

$$\cos(\alpha_1) - \cos(\alpha_2) + \cos(\alpha_3) = M,$$

where M corresponds to the desired fundamental RMS voltage

$$V_{1,\text{rms}} = \frac{16}{\sqrt{2}} \text{ V}.$$

Next, the cancellation equations for the 3rd and 5th harmonics:

$$\cos(3\alpha_1) - \cos(3\alpha_2) + \cos(3\alpha_3) = 0,$$

$$\cos(5\alpha_1) - \cos(5\alpha_2) + \cos(5\alpha_3) = 0.$$

These three nonlinear equations are solved numerically in MATLAB to obtain the switching angles:

$$\alpha_1 \approx 31.42^\circ, \quad \alpha_2 \approx 54.57^\circ, \quad \alpha_3 \approx 69.23^\circ.$$

These angles are then converted into time instants for a 50 Hz period and used to construct the gating patterns in Simulink.

4.1 MATLAB Implementation of SHE Switching Angle Calculation

The following MATLAB script implements the Newton–Raphson method to determine the switching angles α_1 , α_2 , and α_3 that satisfy the Selective Harmonic Elimination (SHE) equations for eliminating the 3rd and 5th harmonics while achieving the required fundamental component.

Listing 1: MATLAB Code Used to Solve the SHE Nonlinear Equations

```

1 % Given parameters
2 V_DC = 20;
3 V1 = 16;
4 target = pi * V1 / (4 * V_DC); % Target for fundamental
5
6 % Initial guesses for alpha1, alpha2, alpha3
7 alpha = [pi/6; pi/3; pi/2];
8
9 % System of nonlinear equations
10 F = @(alpha) [
11     cos(alpha(1)) - cos(alpha(2)) + cos(alpha(3)) - target;
12     cos(3*alpha(1)) - cos(3*alpha(2)) + cos(3*alpha(3));
13     cos(5*alpha(1)) - cos(5*alpha(2)) + cos(5*alpha(3))
14 ];
15

```

```

16 % Jacobian matrix
17 J = @(alpha) [
18     -sin(alpha(1))      sin(alpha(2))      -sin(alpha(3));
19     -3*sin(3*alpha(1))  3*sin(3*alpha(2))  -3*sin(3*alpha(3));
20     -5*sin(5*alpha(1))  5*sin(5*alpha(2))  -5*sin(5*alpha(3))
21 ];
22
23 % Newton-Raphson method
24 tolerance = 1e-6;
25 max_iterations = 100;
26
27 for iter = 1:max_iterations
28     F_val = F(alpha);
29     J_val = J(alpha);
30
31     delta = -J_val \ F_val;
32     alpha = alpha + delta;
33
34     if norm(delta) < tolerance
35         fprintf('Converged in %d iterations.\n', iter);
36         break;
37     end
38 end
39
40 % Display results
41 disp('Alpha values (radians):');
42 disp(alpha);
43
44 alpha_deg = rad2deg(alpha);
45 disp('Alpha values (degrees):');
46 disp(alpha_deg);
47
48 % Convert degree angles into switching instants for 50 Hz
49 alpha1 = 31.4202 * 0.02 / 360;
50 alpha2 = 54.5694 * 0.02 / 360;
51 alpha3 = 69.2269 * 0.02 / 360;
52
53 disp(alpha1);
54 disp(alpha2);

```

55 `disp(alpha3);`

4.2 Numerical Results

The Newton–Raphson algorithm successfully converged to the correct switching angles within five iterations. The obtained values in both radians and degrees are summarized in the following output.

MATLAB Solver Output

Converged in 5 iterations.

Alpha values (radians): 0.5484 0.9524 1.2082

Alpha values (degrees): 31.4202 54.5694 69.2269

Switching instants (seconds): 0.0017 0.0030 0.0038

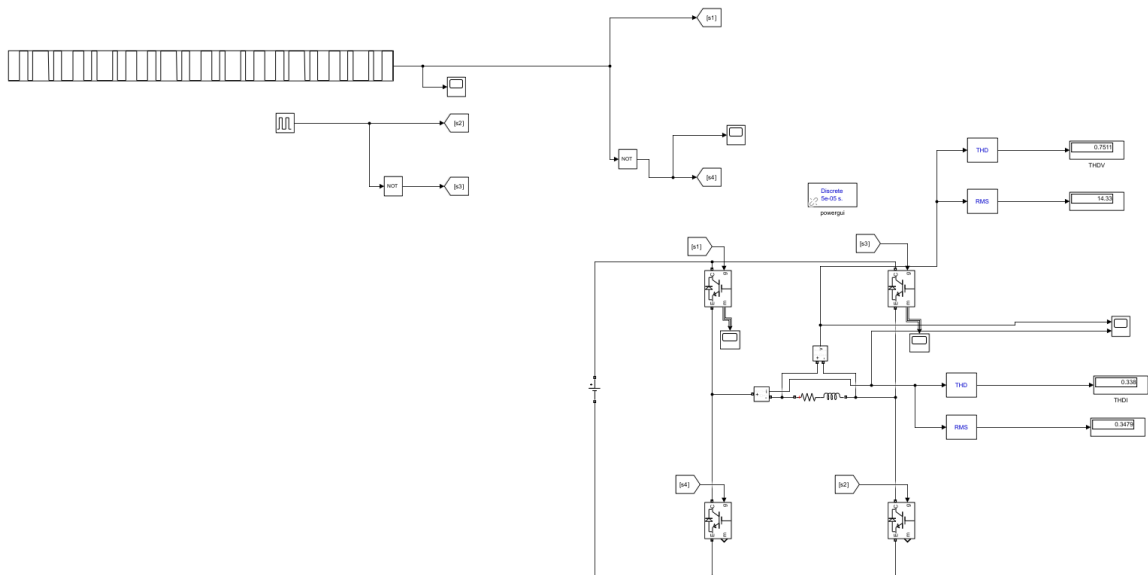


Figure 13: SHE MODEL

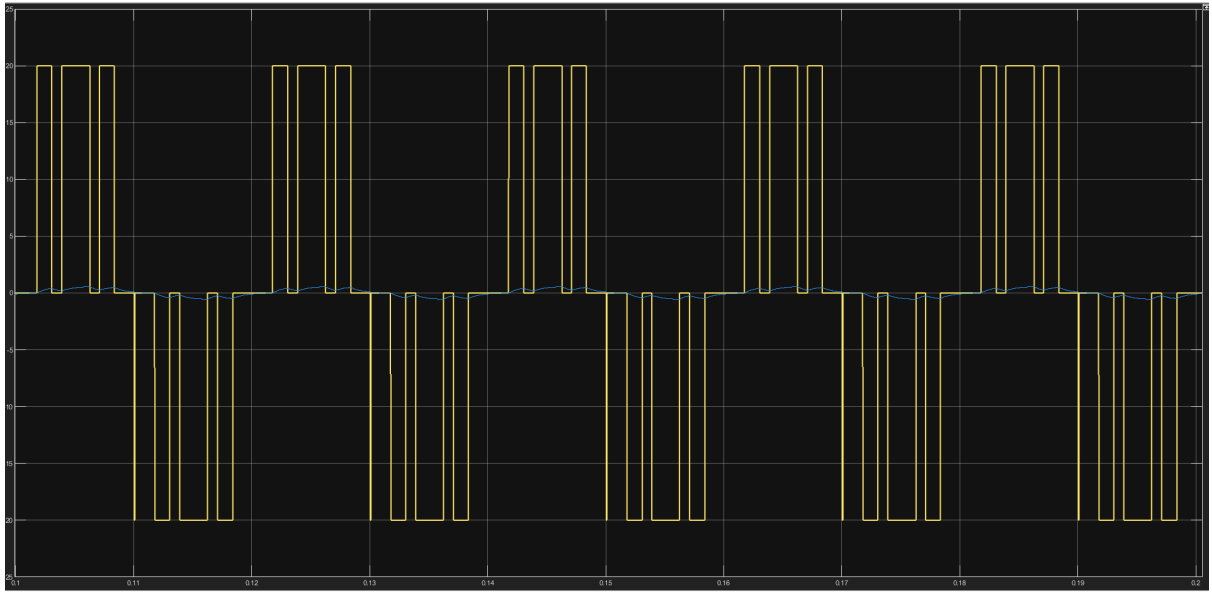


Figure 14: SHE WAVEFORMS

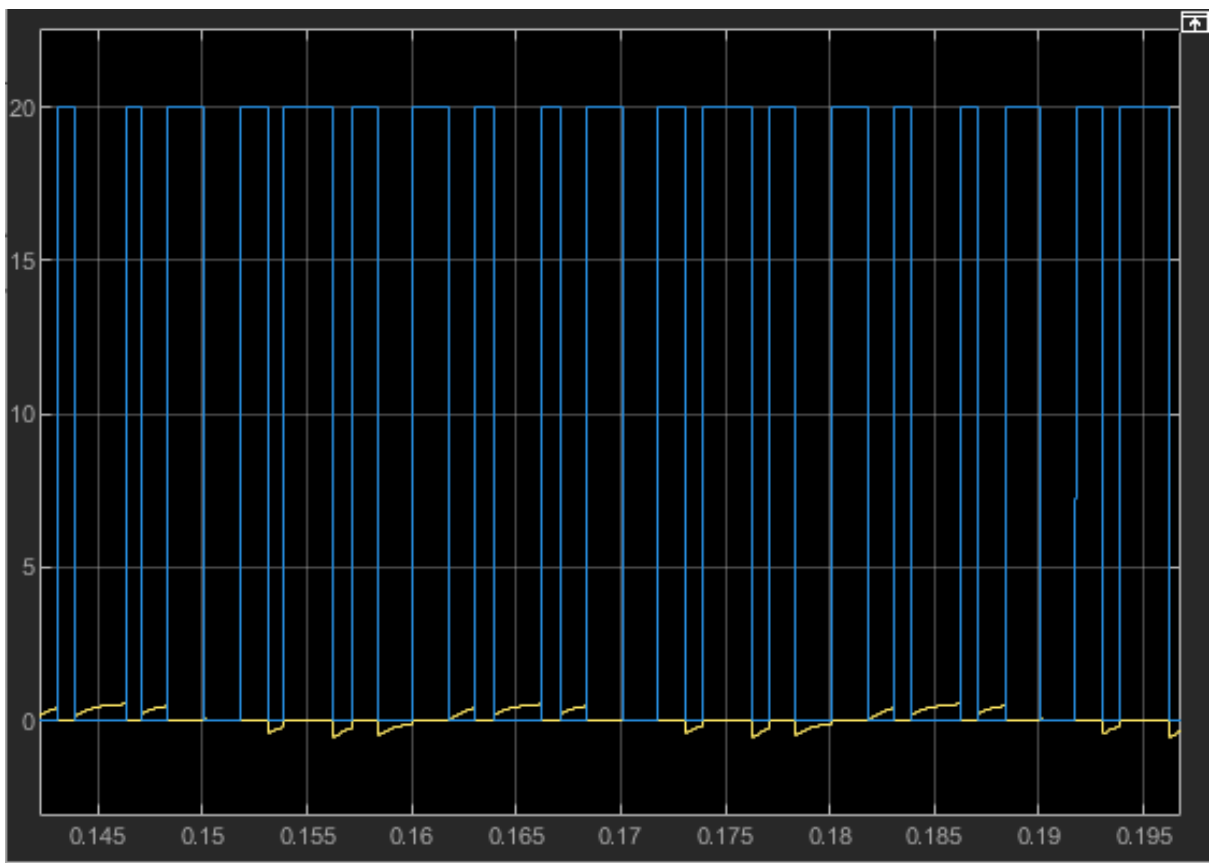


Figure 15: s1,s2

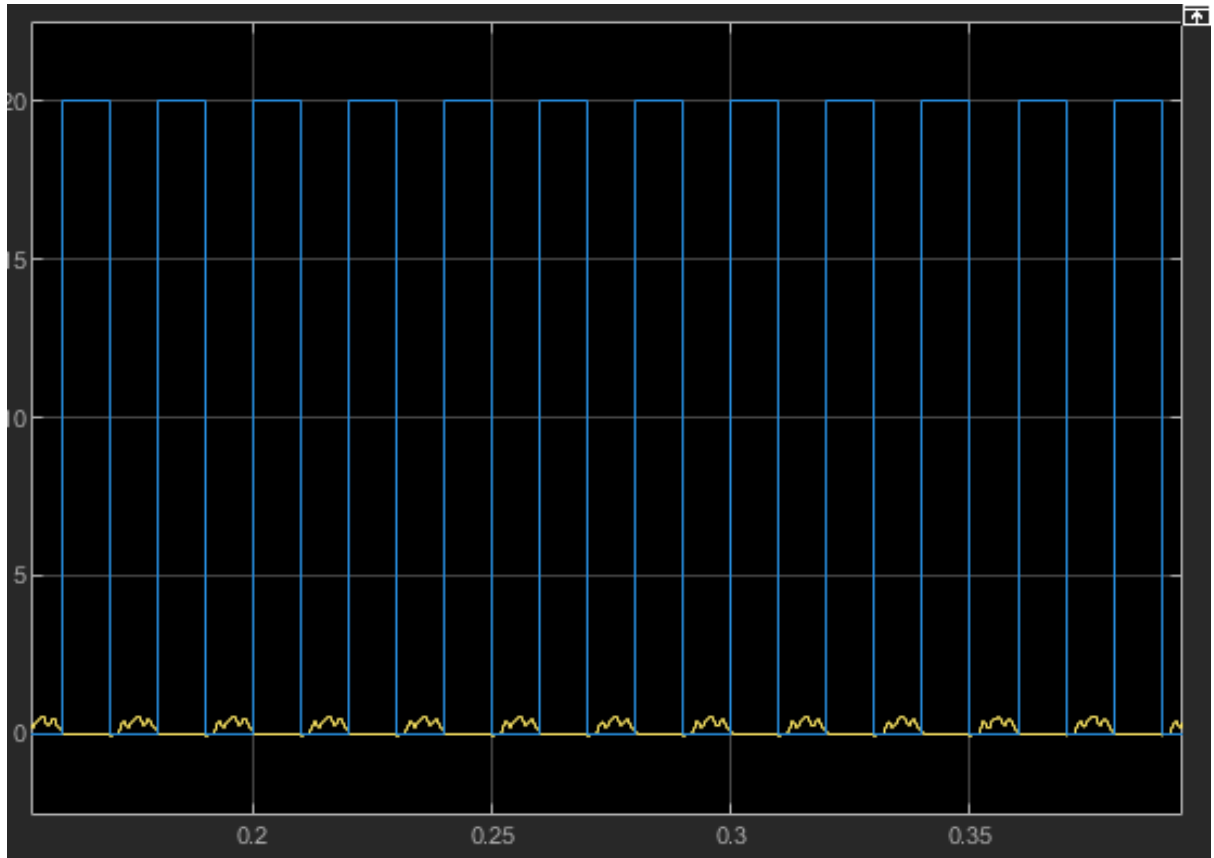


Figure 16: s_3, s_4

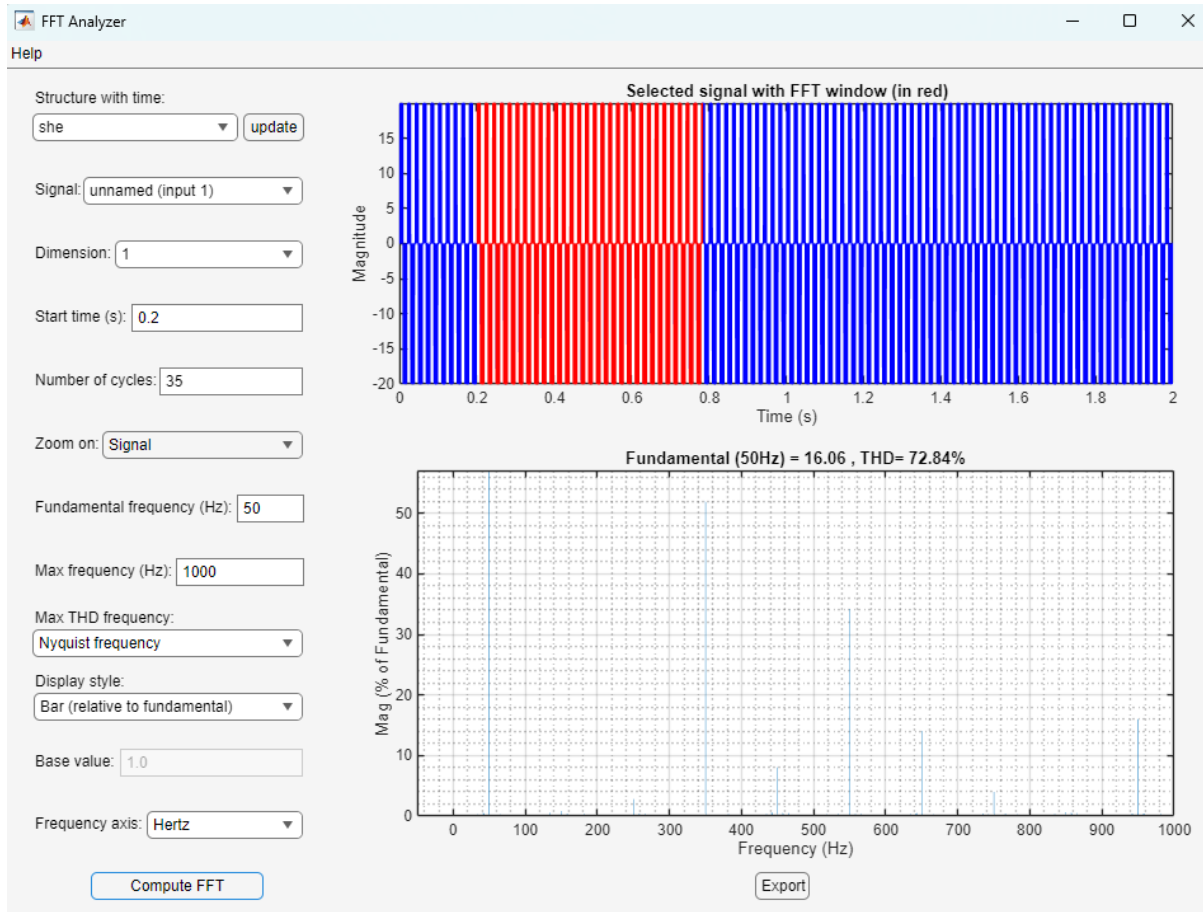


Figure 17: FFT voltage

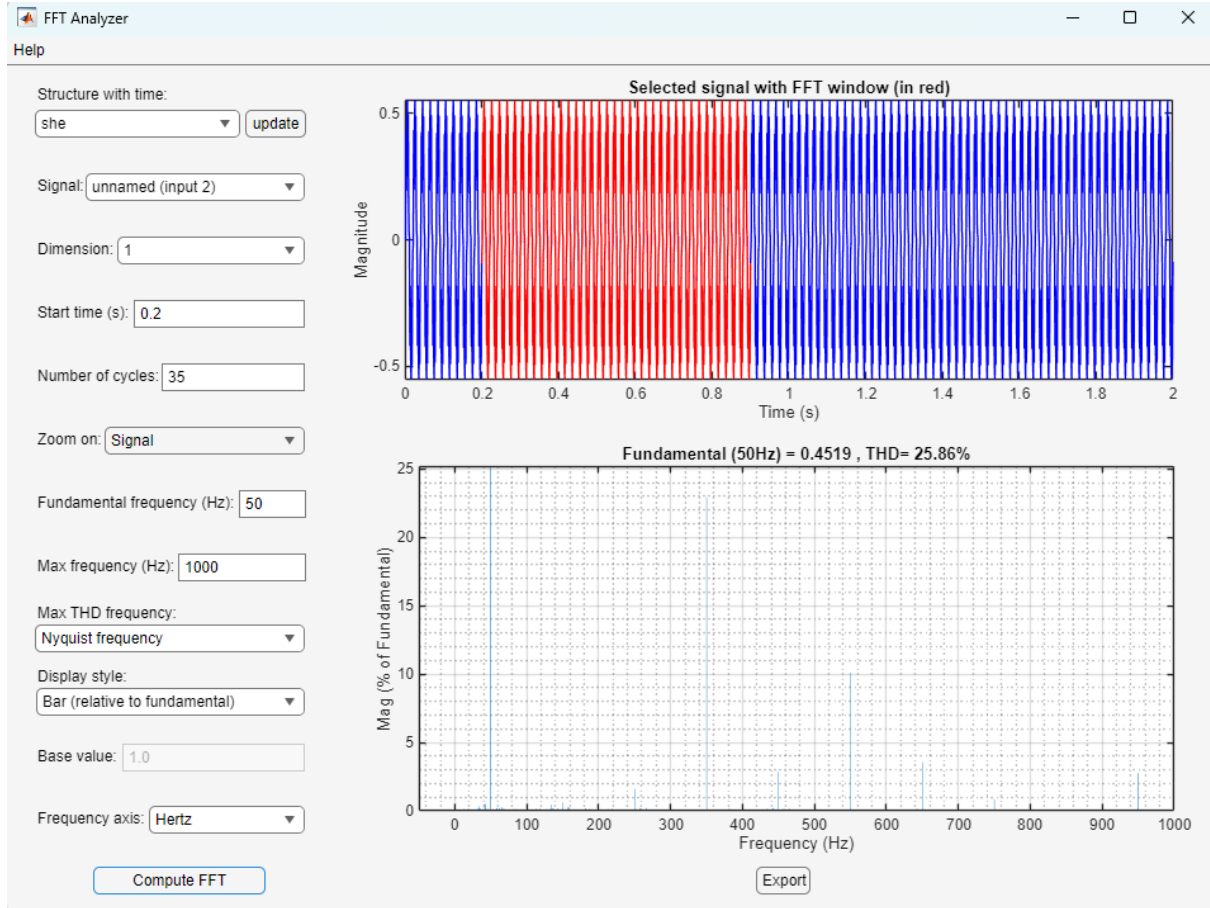


Figure 18: FFT current

5. Bipolar and Unipolar PWM Modulation Techniques

Pulse Width Modulation (PWM) is a fundamental control strategy in power electronic converters, allowing accurate regulation of the output voltage, waveform shape, and fundamental frequency. Among the various PWM approaches, the **bipolar** and **unipolar** modulation schemes are widely implemented in single-phase inverters due to their simplicity and effectiveness.

5.1 Unipolar PWM

In **unipolar PWM**, the inverter output voltage transitions among three distinct levels: $+V_{dc}$, 0, and $-V_{dc}$. This is achieved by modulating the switching devices such that each half-bridge leg is controlled independently. The presence of the intermediate zero level reduces switching losses and significantly suppresses low-order harmonics. Consequently, unipolar PWM generally provides improved waveform quality and lower total harmonic distortion (THD), making it suitable for applications requiring high efficiency and reduced electromagnetic interference.

5.2 Bipolar PWM Simulation Overview

In bipolar PWM, a low-frequency sinusoidal reference and a high-frequency triangular carrier are compared to generate the gate pulses. In this project, the reference is a 50 Hz sine wave and the carrier is a 10 kHz triangular wave, giving a frequency modulation ratio

$$m_f = \frac{f_{\text{carrier}}}{f_{\text{reference}}} = \frac{10,000}{50} = 200.$$

$$f_{\text{carrier}} = 200 \times 50 = 10,000 \text{ Hz}.$$

At every instant, the comparator checks whether the sine reference is greater or smaller than the triangular carrier:

- If $v_{\text{ref}}(t) > v_{\text{tri}}(t)$:
 - Switches S_1 and S_4 are turned ON,
 - Switches S_2 and S_3 are turned OFF,
 - The output line voltage is $v_{AB} = +V_{dc}$.
- If $v_{\text{ref}}(t) < v_{\text{tri}}(t)$:
 - Switches S_2 and S_3 are turned ON,
 - Switches S_1 and S_4 are turned OFF,
 - The output line voltage is $v_{AB} = -V_{dc}$.

Thus, the line voltage v_{AB} switches directly between $+V_{dc}$ and $-V_{dc}$ at the carrier frequency, while its average value over each carrier period follows the 50 Hz sinusoidal reference.

5.2.1 Bipolar PWM Simulation:

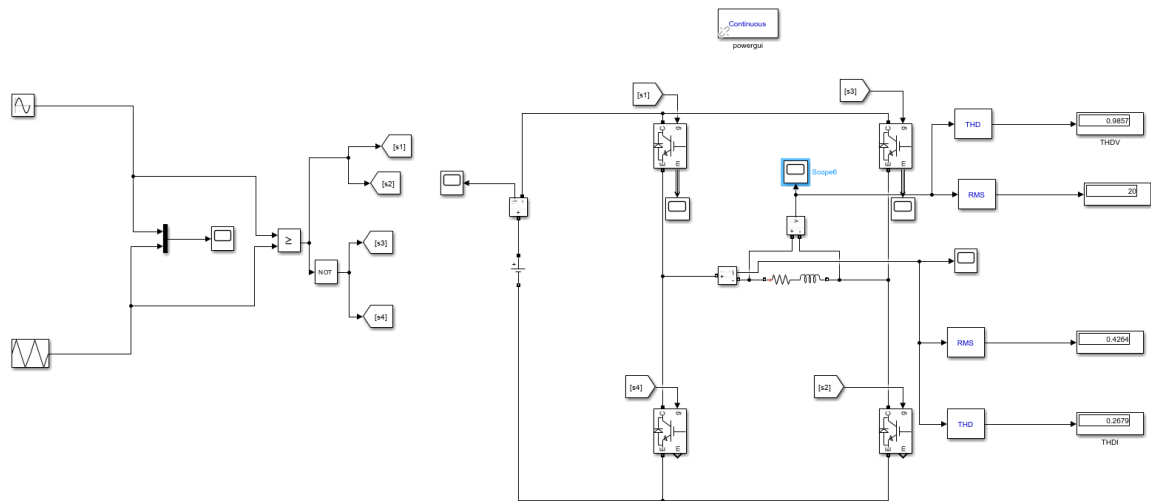


Figure 19: BIPOLAR MODEL

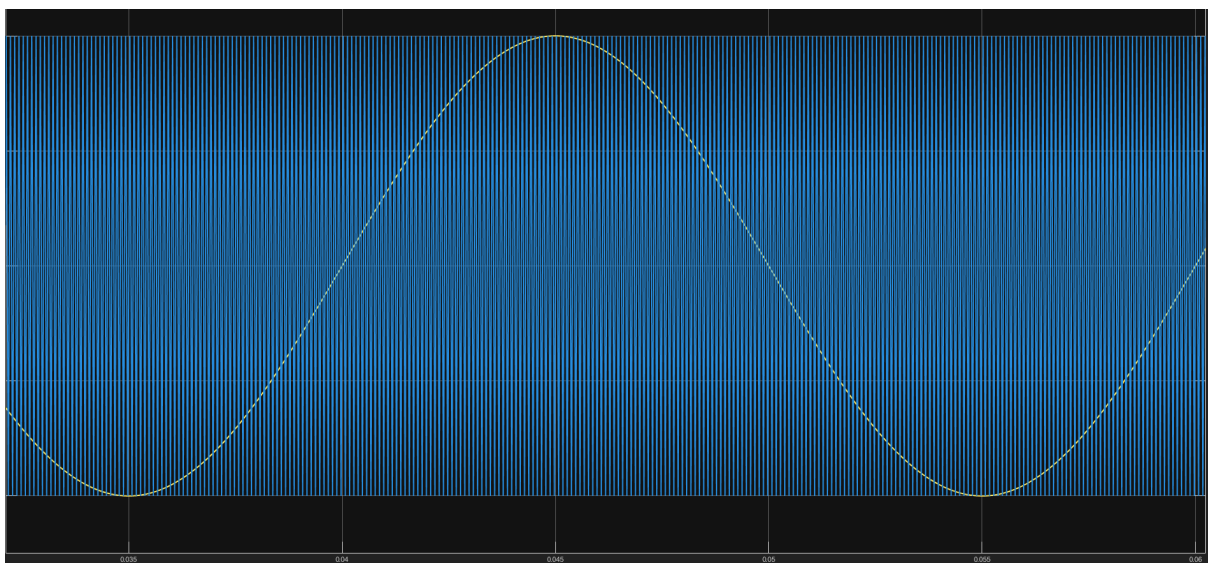


Figure 20: triangular wave & sine wave

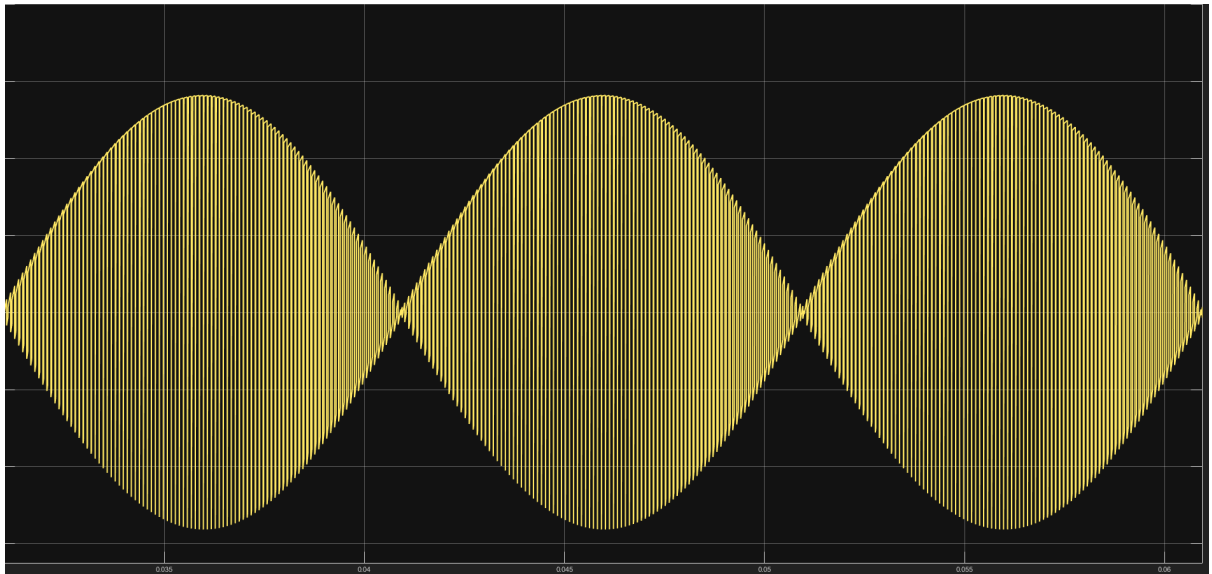


Figure 21: input current

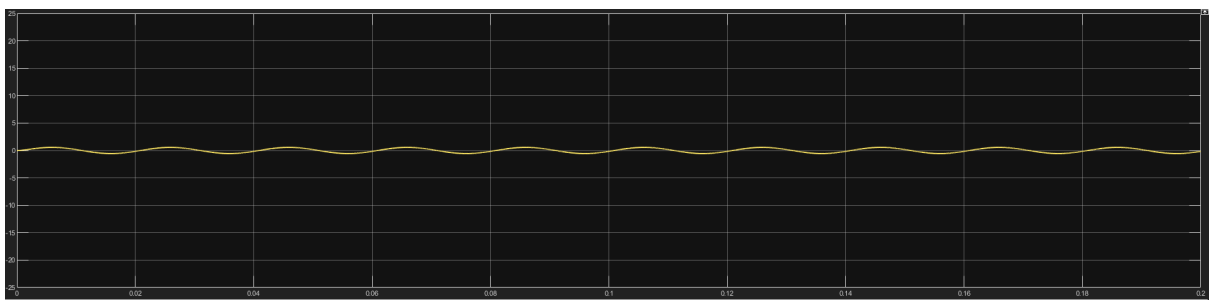


Figure 22: output current

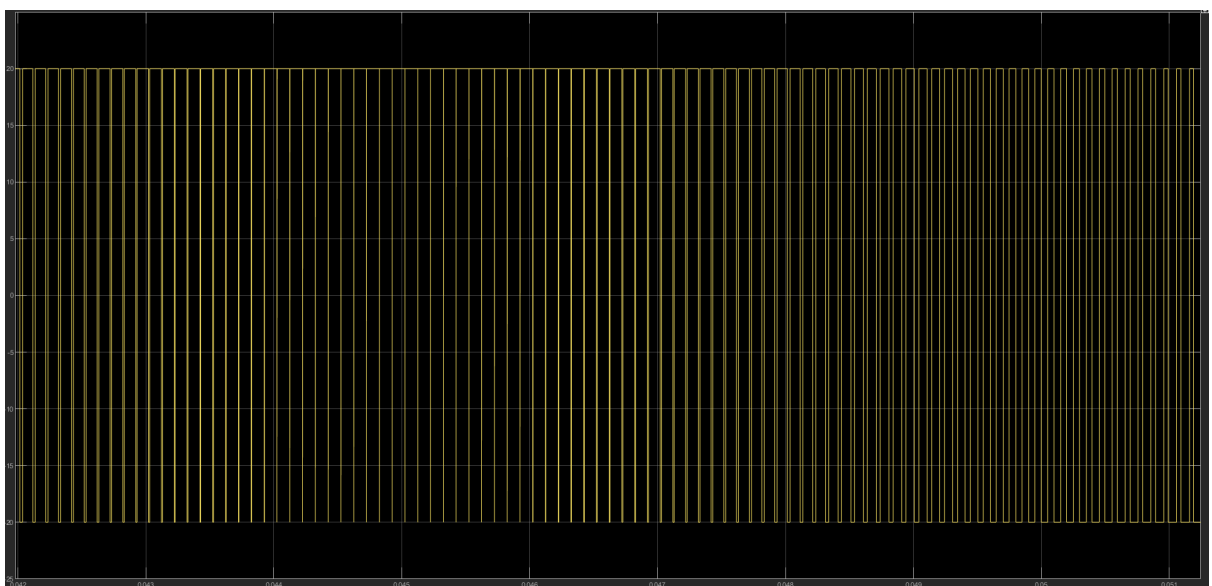


Figure 23: output voltage

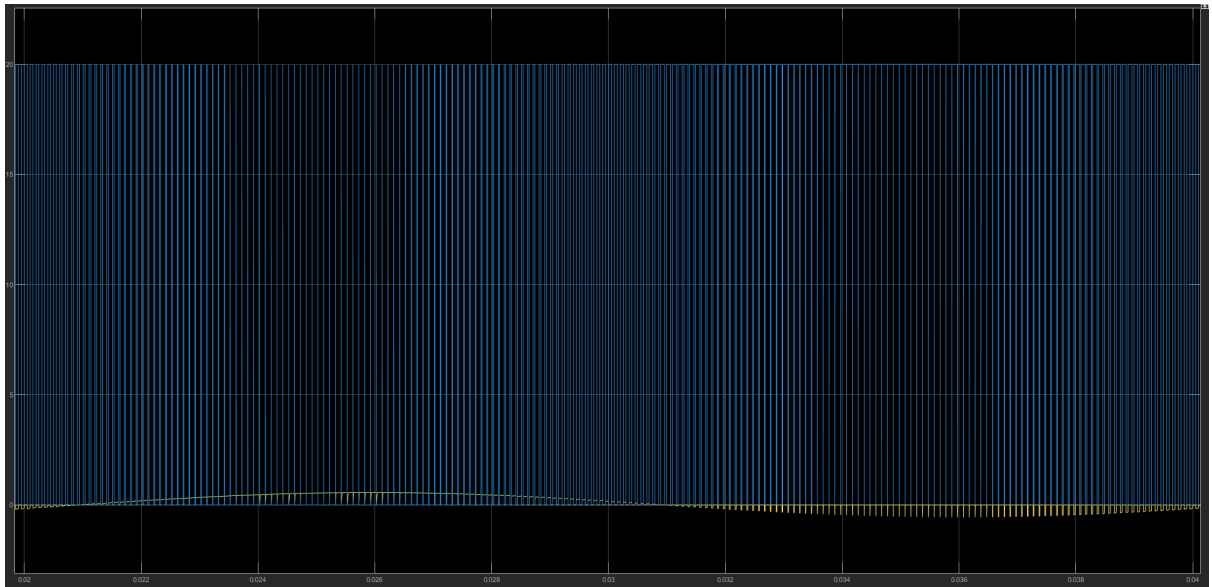


Figure 24: s1,s2

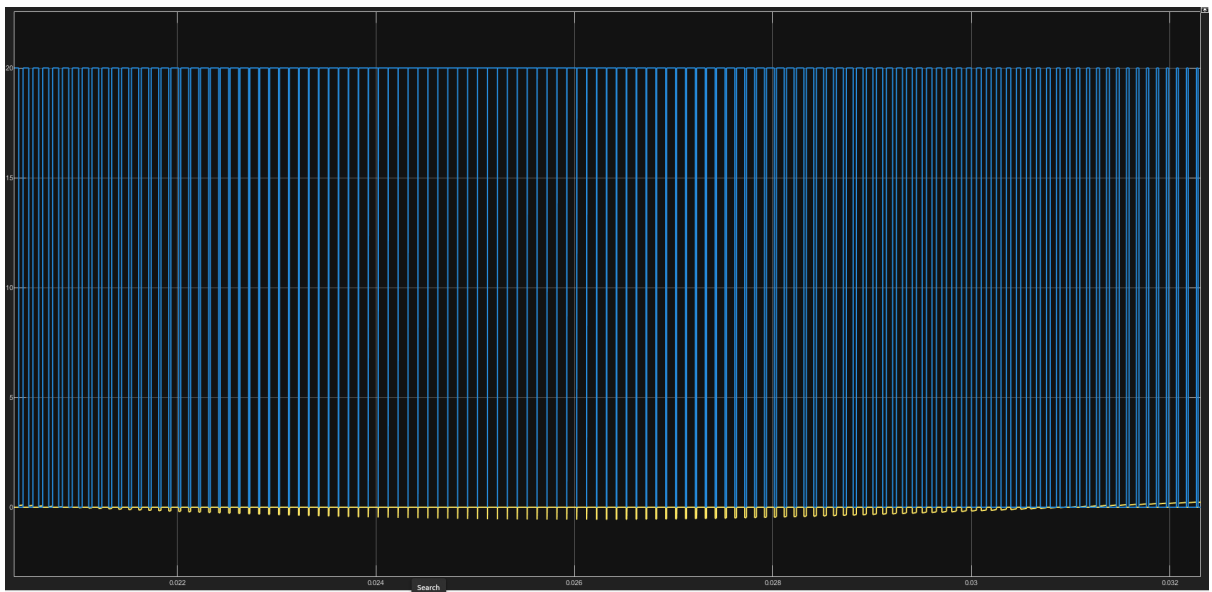


Figure 25: s3,s4

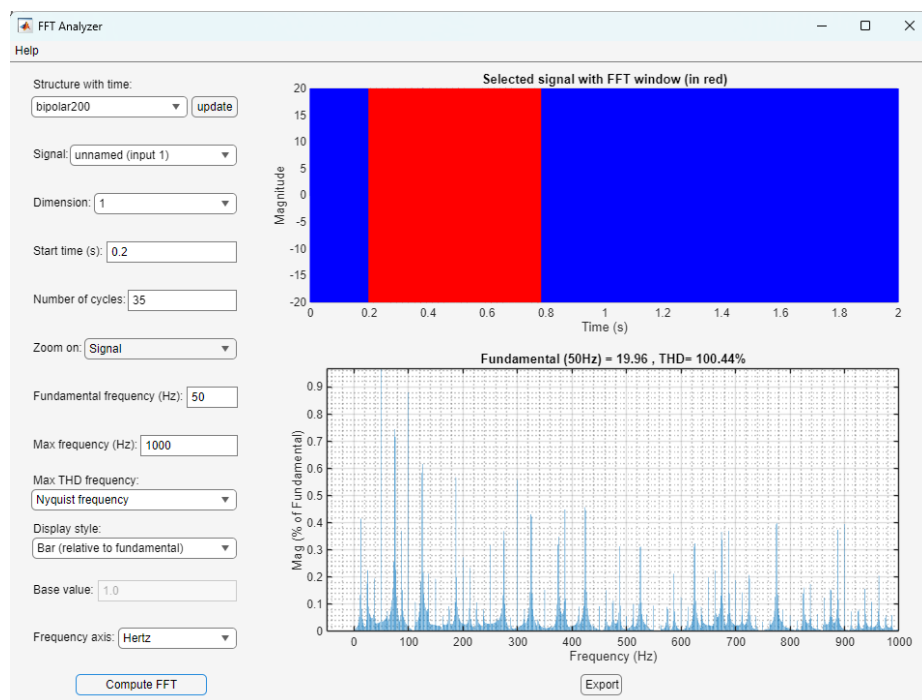


Figure 26: FFT voltage

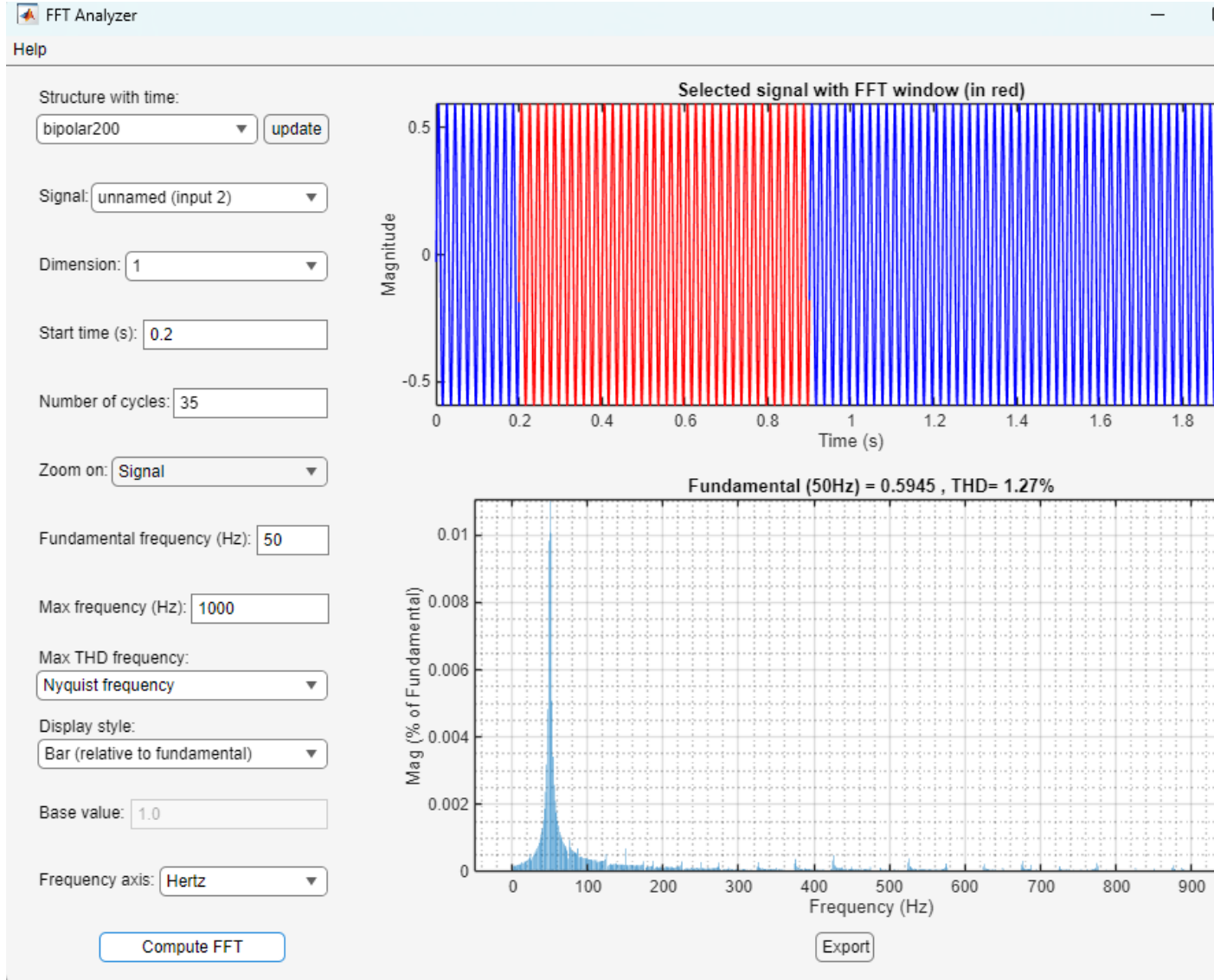


Figure 27: FFT current

5.3 Unipolar PWM Simulation Overview

In unipolar PWM, each leg of the full-bridge inverter is modulated independently using a sinusoidal reference compared with a high-frequency triangular carrier. In this project, two 50 Hz sinusoidal references are used: $v_{\text{ref,A}}(t)$ for the A-leg and $v_{\text{ref,B}}(t) = -v_{\text{ref,A}}(t)$ for the B-leg. Both are compared with the same 10 kHz triangular carrier $v_{\text{tri}}(t)$, so that the frequency modulation ratio remains

$$m_f = \frac{f_{\text{carrier}}}{f_{\text{reference}}} = \frac{10,000}{50} = 200.$$

For the A-leg (switches S_1 and S_2):

- If $v_{\text{ref,A}}(t) > v_{\text{tri}}(t)$:

- S_1 is ON, S_2 is OFF,
- The pole voltage is $v_{AN} = +V_{dc}$.
- If $v_{\text{ref},A}(t) < v_{\text{tri}}(t)$:
 - S_1 is OFF, S_2 is ON,
 - The pole voltage is $v_{AN} = 0$.

For the B-leg (switches S_3 and S_4) using the inverted reference $v_{\text{ref},B}(t) = -v_{\text{ref},A}(t)$:

- If $v_{\text{ref},B}(t) > v_{\text{tri}}(t)$:
 - S_3 is ON, S_4 is OFF,
 - The pole voltage is $v_{BN} = +V_{dc}$.
- If $v_{\text{ref},B}(t) < v_{\text{tri}}(t)$:
 - S_3 is OFF, S_4 is ON,
 - The pole voltage is $v_{BN} = 0$.

The line voltage is obtained as $v_{AB} = v_{AN} - v_{BN}$, which now takes three levels $(+V_{dc}, 0, -V_{dc})$. Because each leg commutates between 0 and $+V_{dc}$ and the effective switching in v_{AB} occurs at twice the carrier frequency, unipolar PWM reduces low-order harmonics and typically yields lower THD than bipolar PWM.

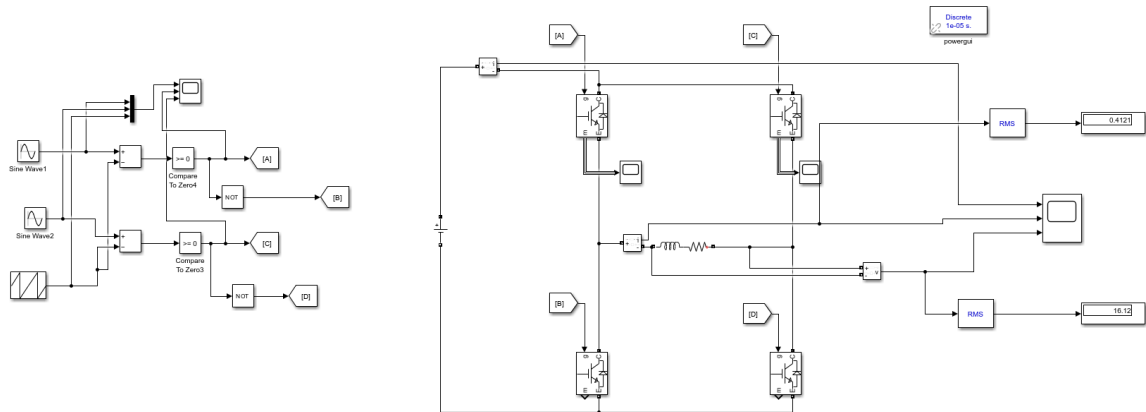


Figure 28: Unipolar MODEL

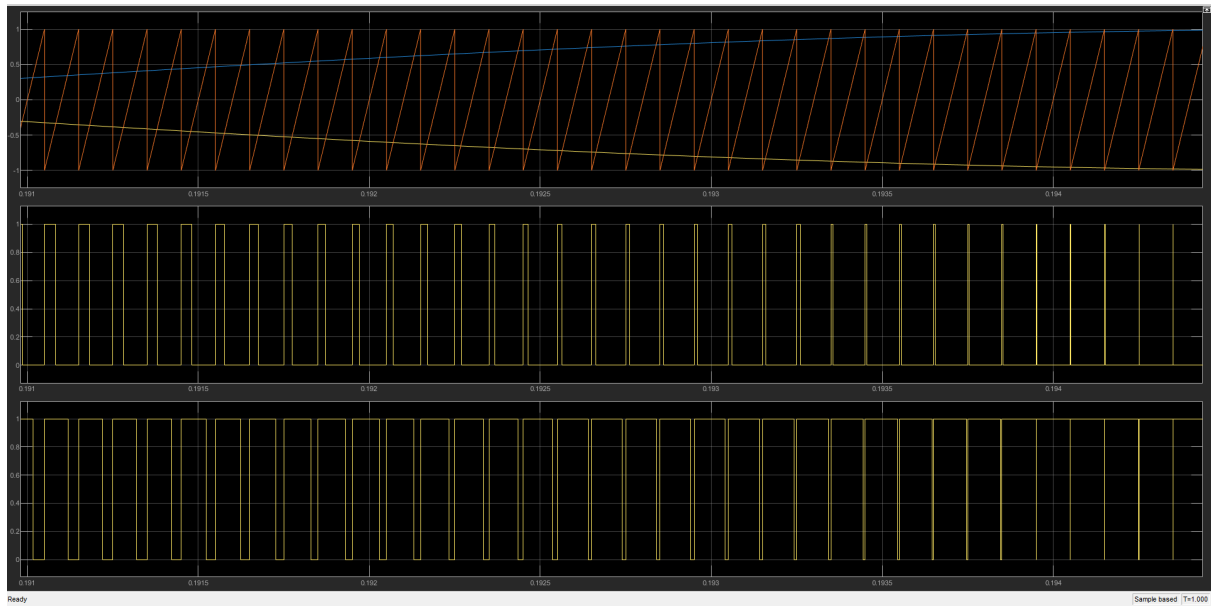


Figure 29: INPUT WAVES

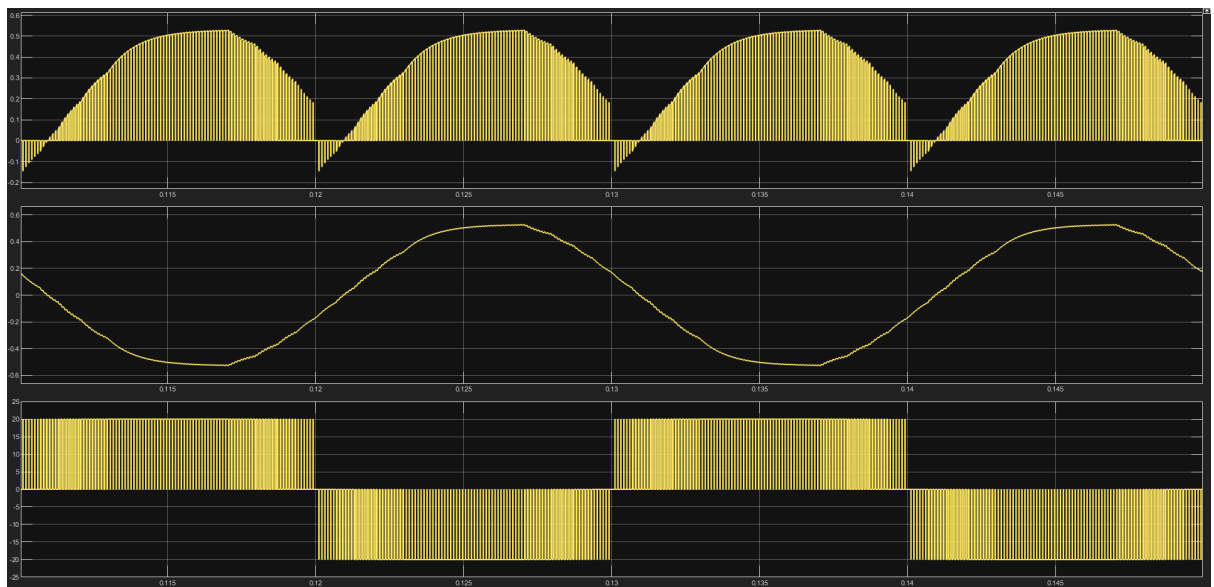


Figure 30: Unipolar WAVEFORMS

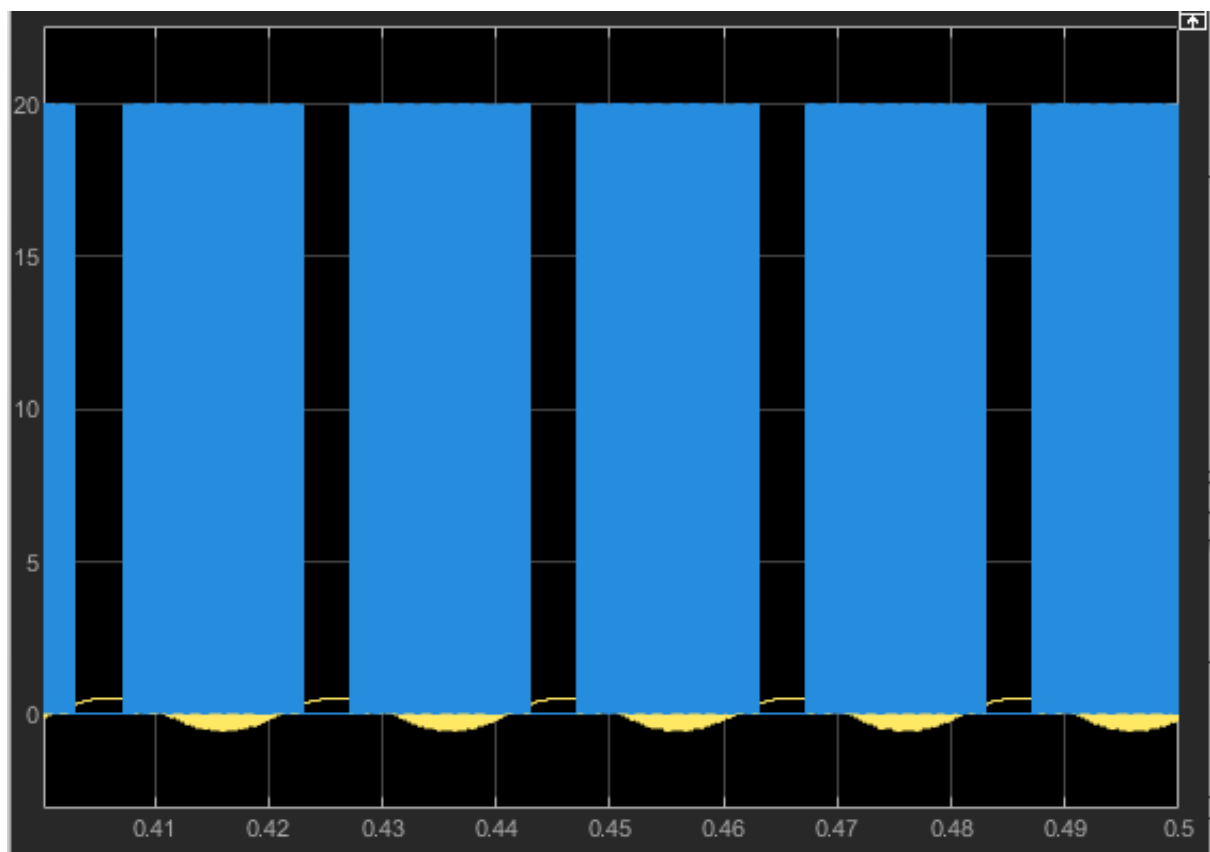


Figure 31: S1,S4

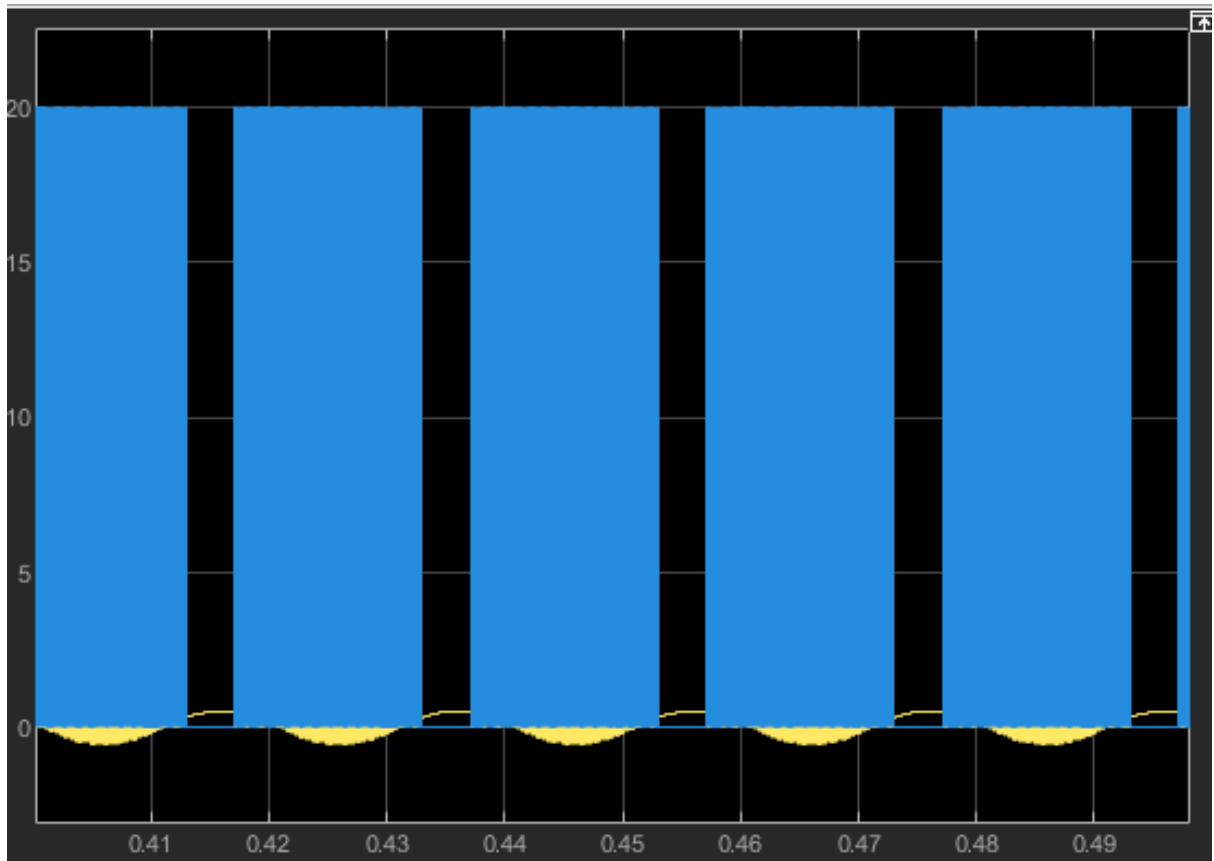


Figure 32: S2, S3

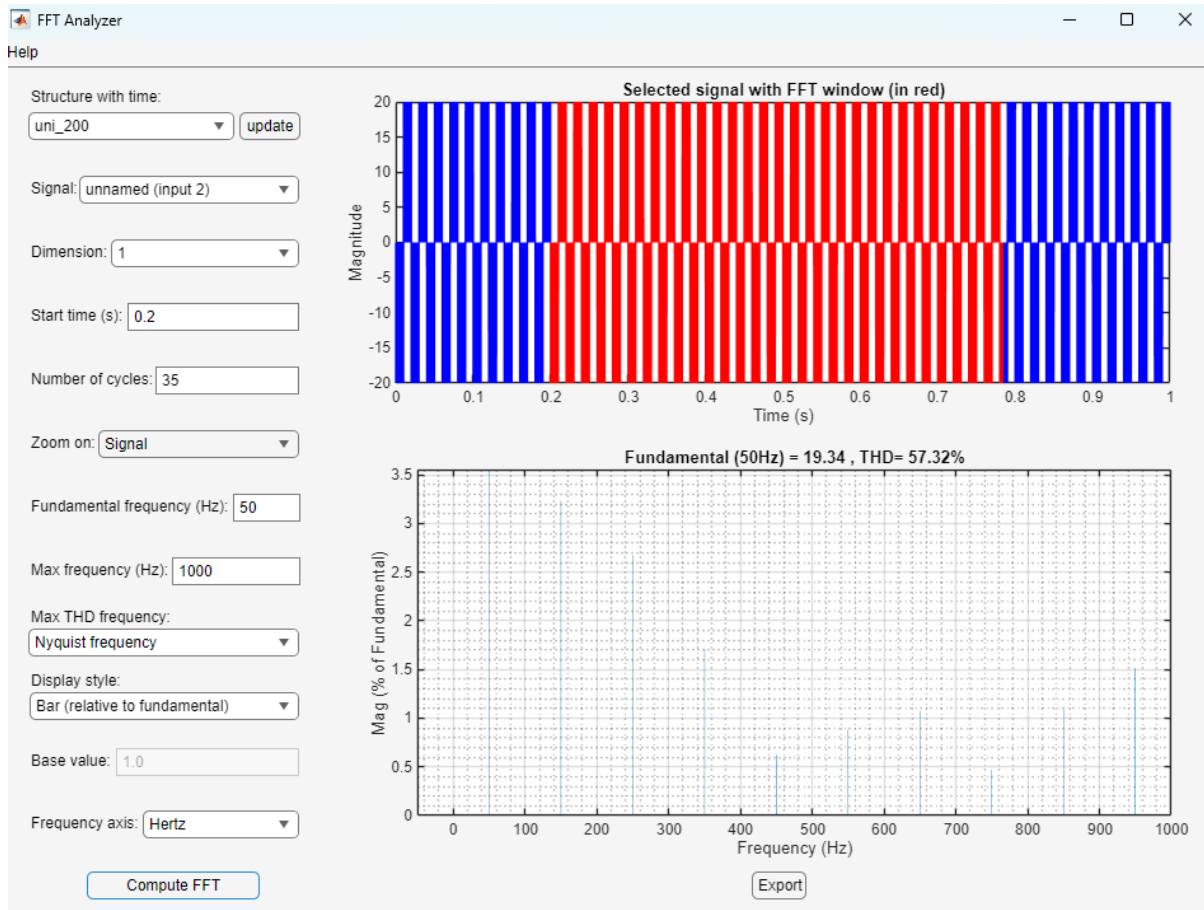


Figure 33: FFT voltage

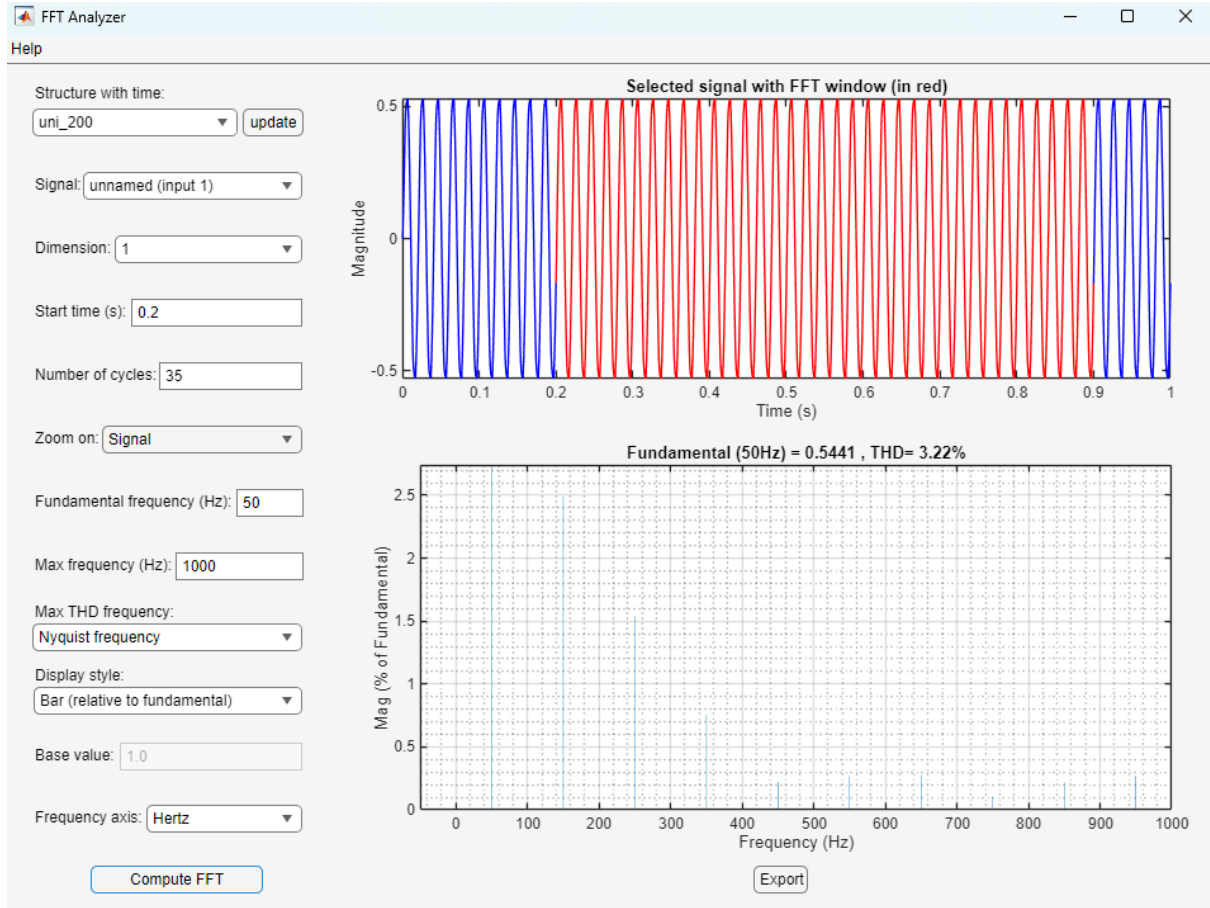


Figure 34: FFT current

6. Bipolar and Unipolar PWM find the minimum value of m_f

In this point, we are required to determine the minimum value of the frequency modulation ratio m_f for both bipolar and unipolar PWM such that the output satisfies

$$V_{1,\text{rms}} = \frac{16}{\sqrt{2}} \quad \text{and} \quad \text{THD}_{\text{current}} \leq 10\%.$$

To achieve this, Python and MATLAB scripts were developed to search for the smallest m_f that meets these conditions. The inverter is supplied by a DC source of $V_{dc} = 20$ V and feeds an R-L load with $R = 34 \, \Omega$ and $L = 33$ mH. The output frequency is fixed at $f_{\text{out}} = 50$ Hz, with corresponding angular frequency $\omega_0 = 100\pi$ rad/s.

In the code, the load impedance at the fundamental is first computed, and the fundamental current and its RMS value are obtained from the specified fundamental voltage. Then, to estimate the current THD, the dominant voltage harmonics are modelled using predefined normalized Fourier coefficients and modulation index values taken from Table 8–3 (Normalized Fourier Coefficients for Bipolar PWM) and Table 8–4 (Fourier Series

Quantities for the PWM Inverter) in *Power Electronics* by D. Hart. For each candidate m_f , the script converts these voltage harmonics into current harmonics through the load impedance, evaluates the RMS sum of the selected harmonics, and calculates the resulting current THD. The minimum m_f that satisfies both the fundamental RMS constraint and $\text{THD} \leq 10\%$ is then identified separately for the bipolar and unipolar PWM schemes.

6.1 unipolar calculations:

Listing 2: MATLAB Code Used to design of unipolar

```

1 clear; clc;
2
3 % Constants
4 Vdc      = 20;
5 R        = 34;
6 L        = 0.033;
7 f_out    = 50;
8 omega_0  = 2*pi*f_out;
9 V1_target = 16;      % fundamental amplitude
10 THD_limit = 10;      % percent
11
12 ma_1 = 0.8;          %ok<NASGU> % not used in
    this code
13 fourier_coeffs = [0.31 0.31 0.14 0.14];
14
15 % Fundamental component
16 V1 = V1_target;
17 Z1 = sqrt(R^2 + (omega_0*L)^2);
18 I1_calculated      = V1 / Z1;
19 I1_calculated_rms  = I1_calculated / sqrt(2);
20
21 % Search variables
22 found = false;
23 max_mf = 5000;
24 min_mf = NaN;
25
26 for mf = 1:max_mf
27     harmonics = [2*mf + 1, 2*mf - 1, 2*mf + 3, 2*mf - 3];
28
29     I_harmonics_rms = zeros(1, numel(harmonics));

```



```

30
31     for k = 1:numel(harmonics)
32         n = harmonics(k);
33         ma_n = fourier_coeffs(k);
34         Vn = ma_n * Vdc;
35         Zn = sqrt(R^2 + (n*omega_0*L)^2);
36         In = Vn / Zn; % amplitude
37         I_harmonics_rms(k) = In / sqrt(2);
38     end
39
40     THD = (sqrt(sum(I_harmonics_rms.^2)) / I1_calculated_rms) *
        100;
41
42     if THD <= THD_limit
43         min_mf = mf;
44         found = true;
45         break;
46     end
47 end
48
49 % Output
50 if found
51     fprintf('Minimum mf = %d, THD = %.2f%%\n', min_mf, THD);
52 else
53     fprintf('No value of mf found that satisfies the THD limit.\n
        ');
54 end

```

MATLAB Solver Output

Minimum mf = 11, THD = 9.33%

$$f_{\text{carrier}} = 11 \times 50 = 550 \text{ Hz.}$$

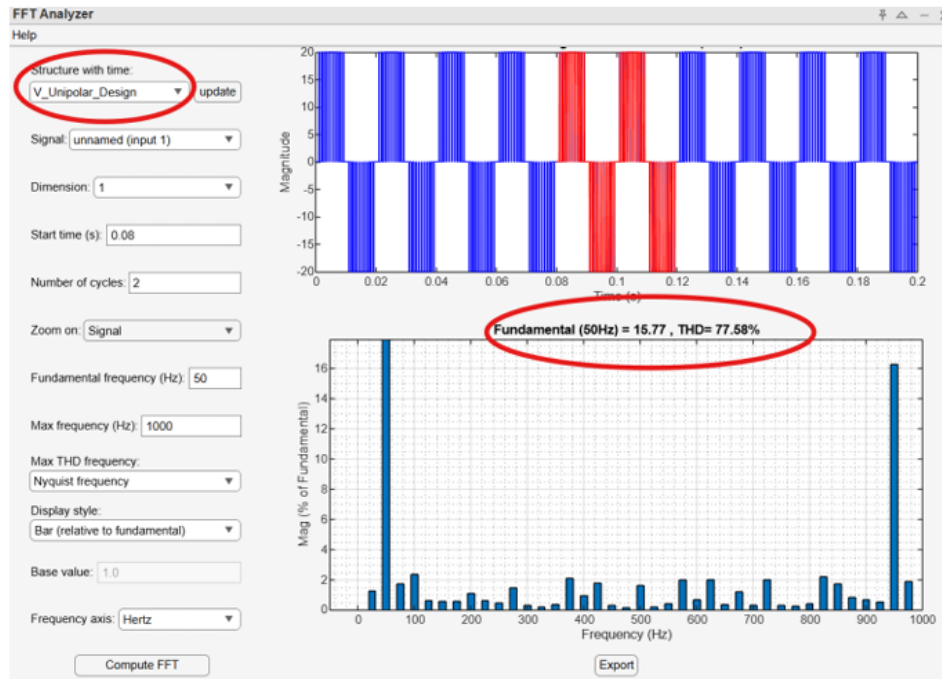


Figure 35: FFT voltage

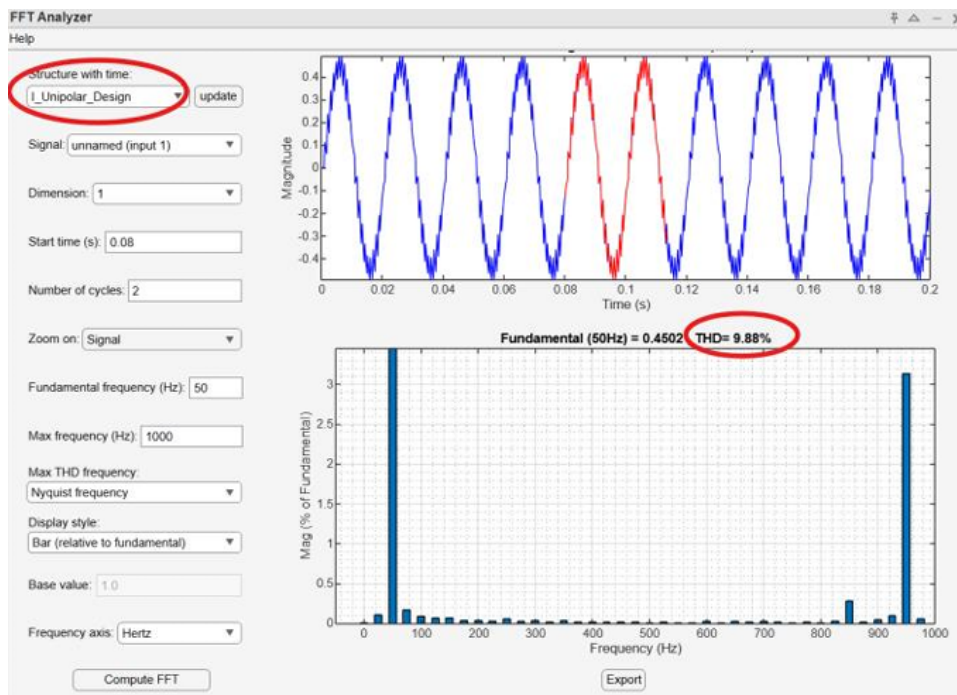


Figure 36: FFT current

6.2 Bipolar calculations:

Listing 3: MATLAB Code Used to design of Bipolar

```
1 clear; clc;
```

```

2
3 % Constants
4 Vdc      = 20;
5 R        = 34;
6 L        = 0.033;
7 f_out    = 50;
8 omega_0  = 2*pi*f_out;
9 V1_target = 16;      % fundamental amplitude
10 THD_limit = 10;      % percent
11
12 ma_1 = 0.8;          %#ok<NASGU> % not used
    explicitly
13 fourier_coeffs = [0.82 0.22 0.22];
14
15 % Fundamental component
16 V1 = V1_target;
17 Z1 = sqrt(R^2 + (omega_0*L)^2);
18 I1_calculated = V1 / Z1;
19 I1_calculated_rms = I1_calculated / sqrt(2);
20
21 % Search variables
22 found = false;
23 max_mf = 5000;
24 min_mf = NaN;
25
26 for mf = 1:max_mf
27     harmonics = [mf, mf + 2, mf - 2]; % sideband orders
28     I_harmonics_rms = zeros(1, numel(harmonics));
29
30     for k = 1:numel(harmonics)
31         n = harmonics(k);
32         ma_n = fourier_coeffs(k);
33         Vn = ma_n * Vdc;
34         Zn = sqrt(R^2 + (n*omega_0*L)^2);
35         In = Vn / Zn; % amplitude
36         I_harmonics_rms(k) = In / sqrt(2);
37     end
38

```

```

39     THD = (sqrt(sum(I_harmonics_rms.^2)) / I1_calculated_rms) *
40         100;
41
42     if THD <= THD_limit
43         min_mf = mf;
44         found = true;
45         break;
46     end
47
48 if found
49     fprintf('Minimum mf = %d, THD = %.2f%%%\n', min_mf, THD);
50 else
51     fprintf('No value of mf found that satisfies the THD limit.\n
52         ');
53 end

```

MATLAB Solver Output

Minimum mf = 38, THD = 9.86%

$$f_{\text{carrier}} = 38 \times 50 = 1900 \text{ Hz.}$$

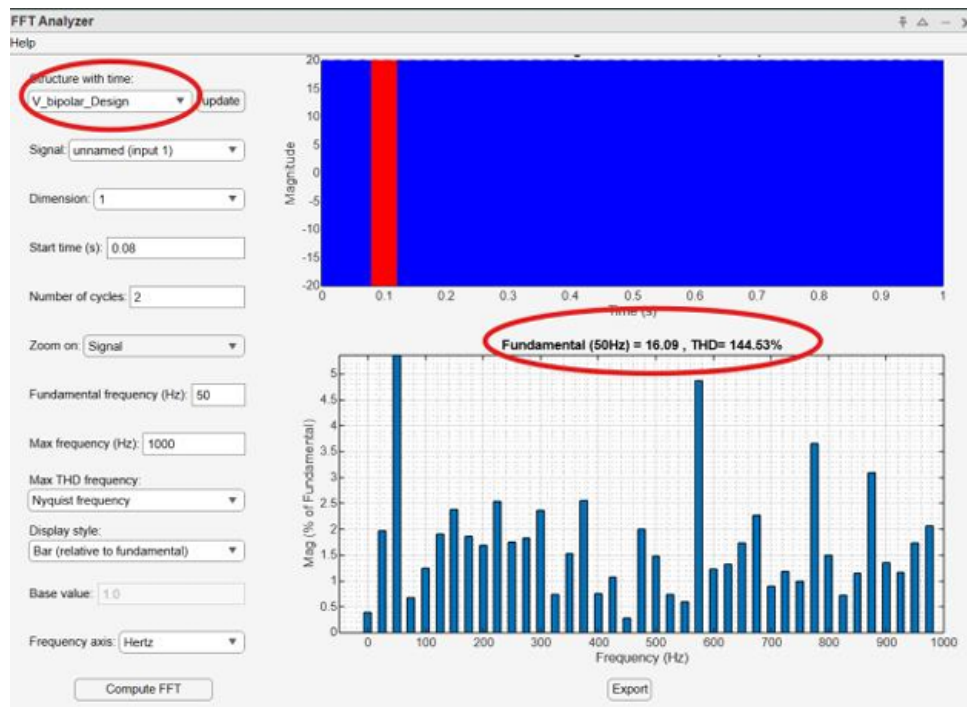


Figure 37: FFT voltage



Figure 38: FFT current

7. Multi-level inverter

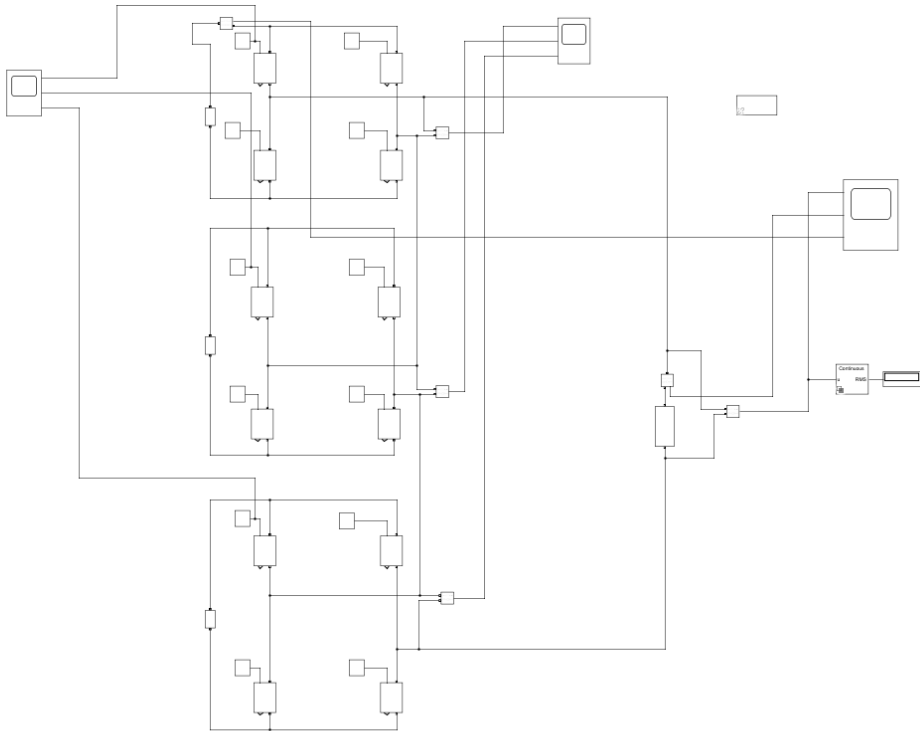


Figure 39: multi level model

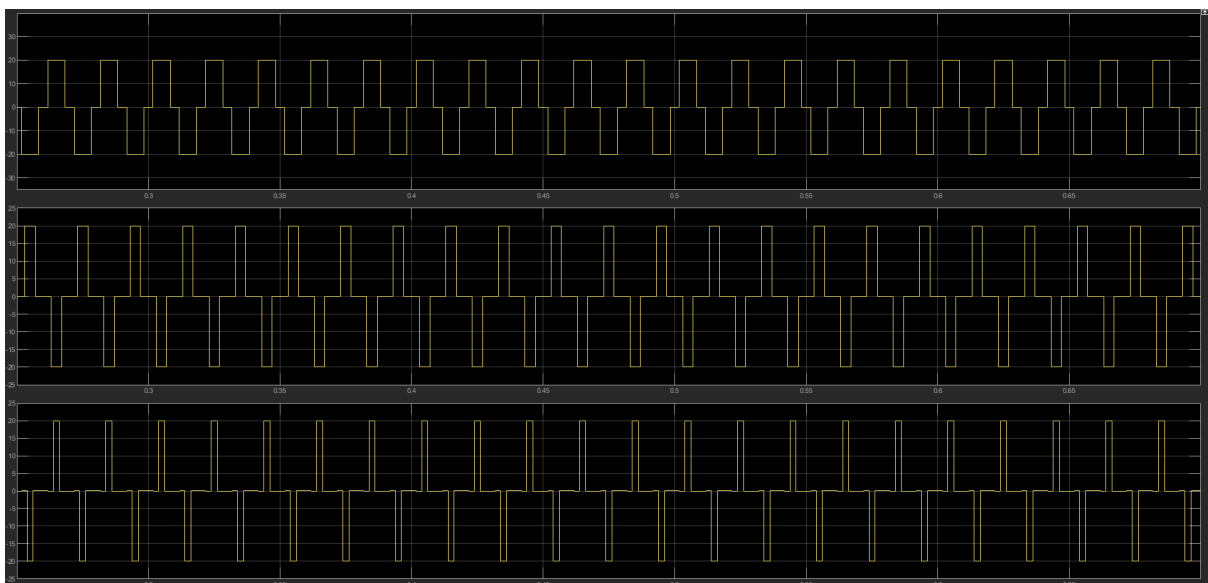


Figure 40: voltage scope

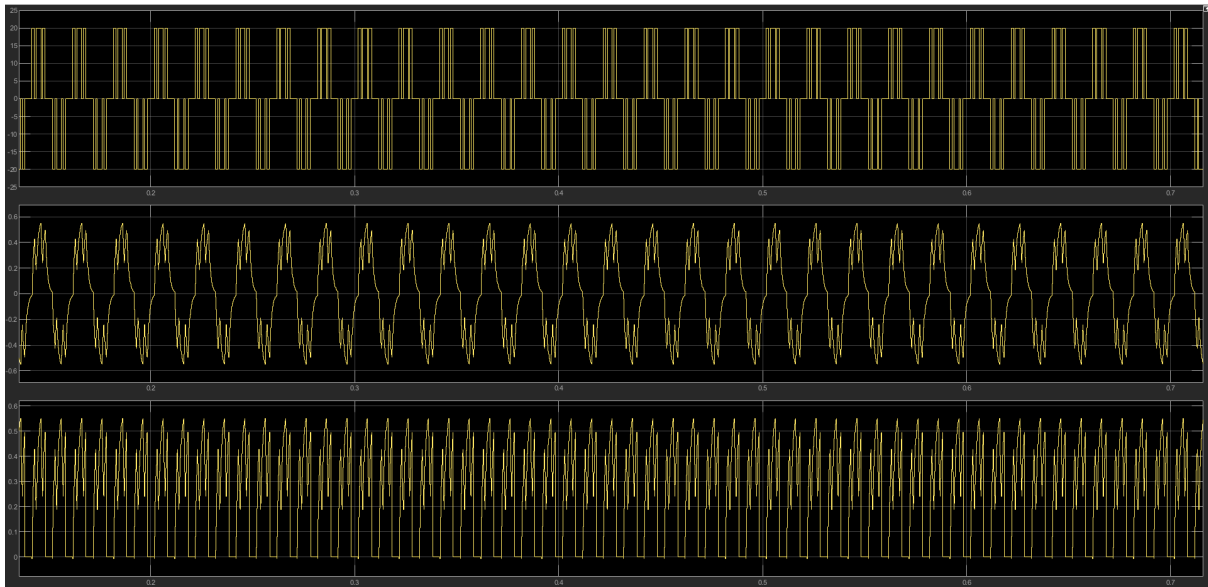


Figure 41: voltage& cuurent scope

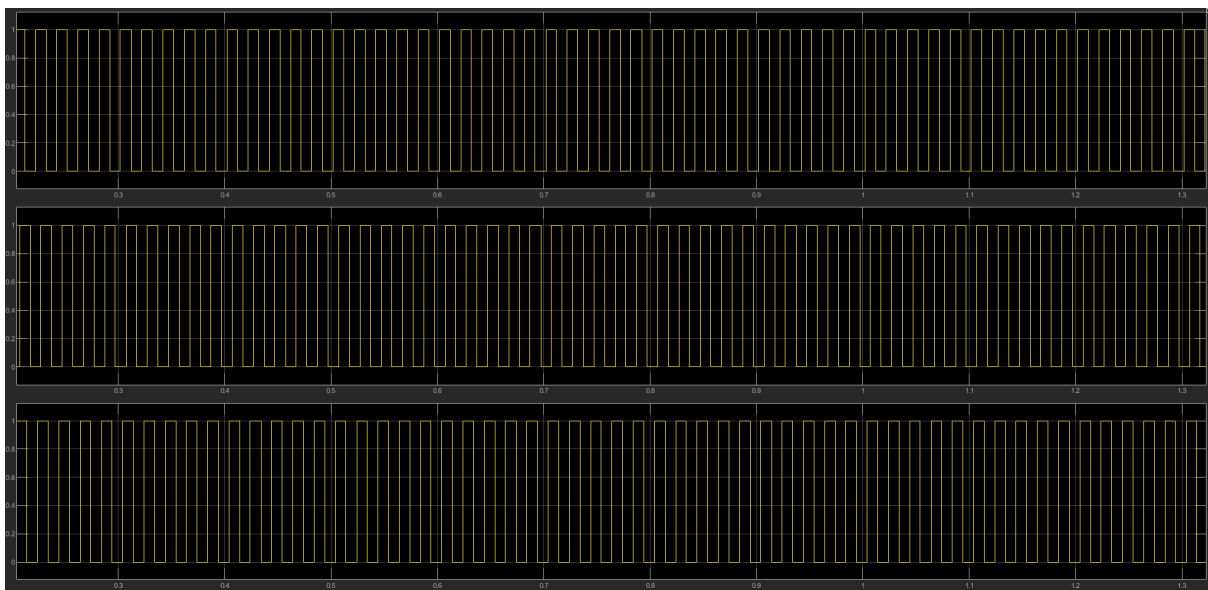


Figure 42: PWM SCOPE

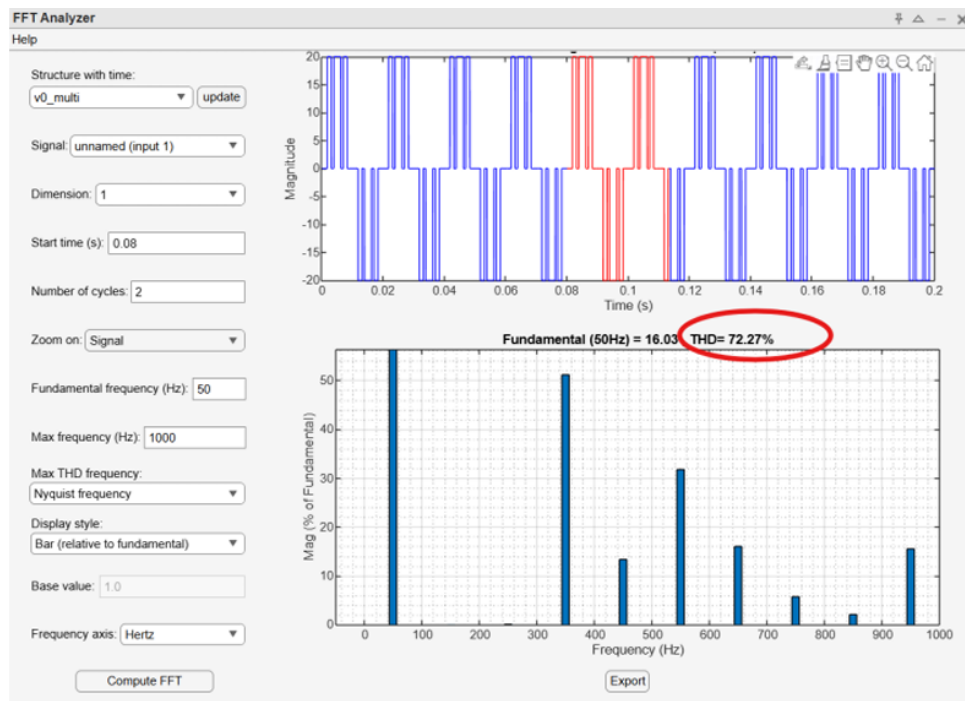


Figure 43: fft voltage

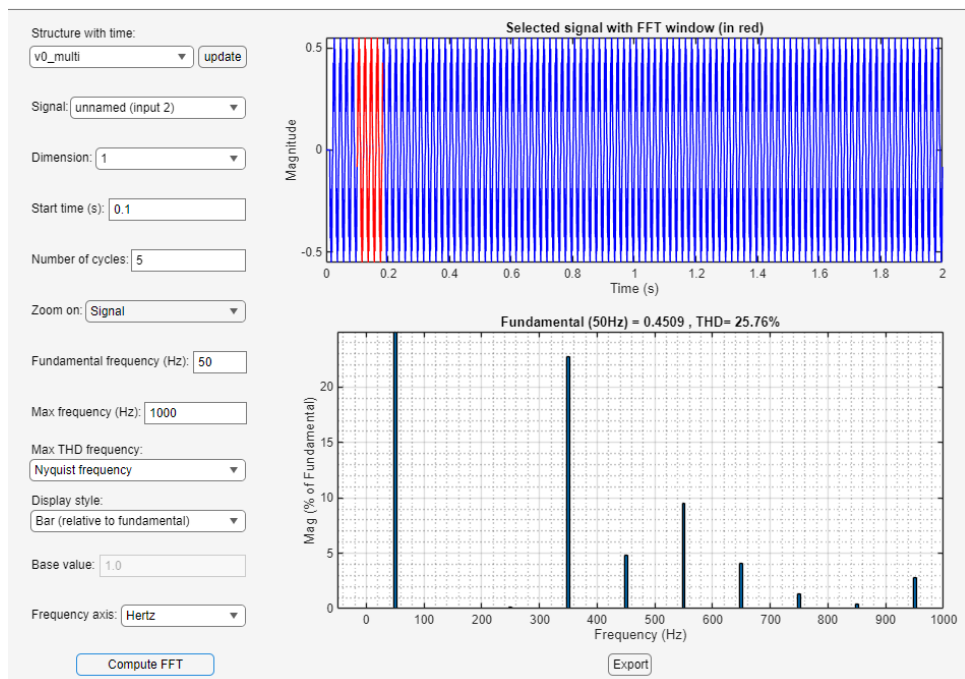


Figure 44: fft current

Table 1: Comparison of Modulation Techniques Based on Fundamental Voltage and Harmonic Distortion

Modulation Technique	V_1 (V)	THD_i (%)	THD_v (%)
Square Wave	25.35	29.02	48.43
Quasi Square Wave	24.6	19.77	31.92
Selected Harmonic Elimination	16.06	25.86	72.84
Multilevel Inverter	16.03	25.76	72.27
Bipolar PWM ($m_f = 200$)	19.98	1.27	100.18
Unipolar PWM ($m_f = 200$)	19.34	3.22	57.132
Bipolar PWM (Design)	16.09	9.76	144.53
Unipolar PWM (Design)	15.77	9.88	77.58